

ADVANTEST®

ADVANTEST CORPORATION

*T5500 Series
Controller Command
Reference Manual*

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*Applicable OS
ASX/U-51*

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Preface

Important Notice

Descriptions about the T5591 in this manual apply to the T5591R unless otherwise stated. Descriptions about the T5581 apply to the T5586, T5585, T5581H and T5581P unless otherwise stated. Descriptions about the T5500 series apply to the T5593, T5592, T5591, T5591R, T5586, T5585, T5581, T5581H, T5581P and T5571P unless otherwise stated.

Related Manuals

ATL-51 Test Plan Program Reference Manual(No. : 8431498)

ATL-51 Pattern Program Reference Manual(No. : 8256292)

ATL-51 Pattern Program Reference Manual (T5376, T5375, T5371, T5336 and T5311)(No. : 8289058)

ATL-51 Socket Program Reference Manual(No. : 8256293)

ATL-51 Scramble Program Reference Manual(No. : 8256294)

T5500 Series Tester Utility Program Reference Manual(No. : 8256295)

ATLworks Tool User's Manual(No. : 8311174)

ASX/U-51 Hybrid Controller User's Manual(No. : 8409598)

ASX/U-51 Hybrid Controller Reference Manual(No. : 8409600)

Error Code Table Reference Manual(No. : 8141091)

XATL/U-51 Cross Compiler User's Manual(No. : 8256237)

ADVANSite Test System Software Installation/System Start User's Manual VOL1 Basic OS Installation(No. : 8227728)

ADVANSite Test System Software Installation/System Start User's Manual VOL2 Tester Environment Software Installation(No. : 8227729)

T5500 Series Memory Test System Control Box/Trigger Panel User's Manual(No. : 8256298)

T5300 Series Flash Memory Parallel Measurement Function Description(No. : 8239605)

T5500 Series Memory Test System Calibration Program Reference Manual(No. : 8256296)

T5500 Series Memory Test System Precision Calibration Program Reference Manual(No. : 8311137)

T5500 Series Memory Test System Precision Calibration2 Program Reference Manual(No. : 8350393)

1. Introduction

This chapter explains the controller command outline and individual controller command functions.

1. 1 Controller Commands

Device test programs are executed in the controller mode (tester driver). A variety of service programs used in this mode are divided into utility programs and other programs activated by key commands.

This manual explains the operation of programs activated by these key commands (controller commands). This manual also explains the key commands having functions equivalent to those of the control box keys.

➔ For more information on the utility program, refer to [T5500 Series Tester Utility Program Reference Manual\(No. : 8256295\)](#).

1. 2 List of Controller Commands

1. 2. 1 Key Commands



Tip

S, I, Z, V, and .A commands have some restrictions for the T5500 ADVANsite systems (ASX/U-51).

➔ For more information on respective commands, refer to [2. "Operation" on page 2-1](#).

Commands that start or abort the test program

Space	Starts test program execution.
CTRL/C	Forcibly aborts a running test program.
O	Selects the mode when the name of the program to be used or manual analysis is specified.

Commands that change the I/O unit or specify I/O-related processing

S	Changes the output unit of the WRITE statement and reports.
CTRL/C	Executes form feed operation for the output unit specified by the S command. For example, it clears the screen for VKT, and outputs CR (carriage return) and FF (form feed) for LPT.
I	Changes the input device specified in the INPUT statement.

Commands that select the test program operation mode or execute processing related to data log files

F	Sets the function key mode.
M	Sets the data logging mode.
Z	Sets the action selector mode.
J	Opens a data file.
V	Closes a data file.
.S	Specifies the number of the sample to be added to the data file.
.V	Initializes (INIT) or releases (RELEASE) the customer.
.T	Sets the test fail counter.

Commands that list or clear the counter values

C	Clears the measurement counter values, test fail counter information, and DC parametric test result.
L	Lists the measurement counter values or the test fail counter values.
N	Lists the fail test numbers.

Commands that execute processing related to debugging

A	Selects the disk access check mode to check whether the disk access command has been issued.
T	Specifies the entire trigger required for debugging.
P	Specifies only part of trigger required for debugging.
CTRL/B	Forcibly and temporarily stops running the test program.
CTRL/T	Selects the trace mode to trace the current execution process being carried out by the test program.

Other Commands

U	Clears program areas, closes data files, releases key commands, and changes customers.
CTRL/E or ESC	Passes control from the controller to the file monitor.
/XXXX.␣	Starts a variety of utility programs.
.A	Sends operational management information to the TESTARIUM or ADVANstar-21 service computer.
CTRL/I	Inhibits automatic start of the system initialization program.
CTRL/S	Temporarily stops upward scrolling of the CRT screen.
CTRL/Q	Restarts upward scrolling of the CRT screen. In other words, resumes the operation stopped by CTRL/S.
NO SCROLL (HOLD SCREEN)	Pressing this key once is equivalent to entering the CTRL/S or CTRL/Q command. In other words, CTRL/S and CTRL/Q commands are executed alternately.
CTRL/Z	In response to a query of the INPUT statement, CTRL/Z clears all characters being entered and exits from the INPUT routine.
SHIFT/-	In response to a query of the INPUT statement, SHIFT/- clears all characters being entered and waits for another input.

1. 2. 2 Utility Programs

Tip

The following utility programs are activated and executed by the VKT operation in the controller mode. Some other utility programs may be called by test plan programs.

➔ For more information, refer to [T5500 Series Tester Utility Program Reference Manual\(No. : 8256295\)](#).

1. 2 List of Controller Commands

Activation command	Name	Function
/PM	Pin monitor	Confirms setting conditions.
/unit name	Data set	Changes the unit setting conditions.
/DS	Data scan	Varies the unit setting value in steps.
/DEBUG or CTRL/D	Debugger	Debugs the test plan and subroutine programs.
/SHM2	Two-dimensional shmoo plot	Shmoo plot determined by two parameters.
/SHM3	Three-dimensional shmoo plot	Shmoo plot determined by three parameters.
/FBMAP	Fail-bit map	Creates the fail-bit map of the memory device.
/MPDISP	Parallel test monitor	Displays the parallel test results obtained using the handler or prober.
/HSET, /HD	Header command	Header information setting and output.
/HD1, /HD2	Enlarged character header command	Header information output using enlarged characters.
/MTRACE	Pattern tracer	Performs the tracing debug for each pattern program step.
/MLOAD	----	Loads the specified program into the user area.
/MLIST	----	Displays the user area status.
/BUFF	----	Designates the BUFF (compilation buffer) program name.
/PRO	----	Specifies the execution program.
.J, .H	HIDAR	Sets up HIDAR conditions and outputs the histogram.
/WFMAP	Wafer map	Wafer map output.
/FTDISP /FTLIST	Fail test display	Displays test results.
/YIELD	Yield report	Creates the yield report.
/DTLOG	Data log	Sets the data log mode. (This is equivalent to the M command.)
/PART	----	Executes only the specified test number.
/GLOBAL /SWITCH /HEADER	----	Sets global variables.

1. 3 Command Input

Input method

- For commands designated by one alphabetical character, enter the character. When characters are entered, they are not displayed. However, the corresponding message is displayed.
- For commands beginning with a period (.), be sure to specify a period before letters. A period (.) and letters are not displayed when they are entered. However, the corresponding message is displayed.

Tip

If only a period (.) is entered, the test program may be halted or no other commands may be accepted.

- For commands beginning with a slash mark (/), be sure to specify a slash mark (/) before the command character. The character string terminator is a ↵ (CR: carriage return). The characters between a slash mark (/) and the carriage return (↵) are displayed when they are entered.
 - ➔ For more information on commands beginning with a slash mark (/), refer to [T5500 Series Tester Utility Program Reference Manual\(No. : 8256295\)](#).
- For commands beginning with CTRL, such as the CTRL/C command, enter an alphabetical character while pressing the CTRL key. The input character is not displayed when it is entered. However, the corresponding message is displayed.
Some commands do not display a message.

Input acceptable conditions

- Commands accepted under any conditions:
 - CTRL/C command
 - CTRL/S command
 - CTRL/Q command
- Commands accepted when the RDY (ready) lamp on the control box is lit:
 - All commands except CTRL/B.
- Commands accepted when the TSTG (testing) and the PAUSE lamps on the control box are lit; in other words, when the test program is temporarily halted:
 - All commands except U.
- Commands accepted when the TSTG lamp on the control box is lit and test mode has been set to AUTO:
 - CTRL/C command
 - CTRL/S command
 - CTRL/Q command
 - Space command
 - M command

F command

J command

V command

Z command

L command

N command

O command

H command

CTRL/B command

CTRL/D command

Any command that begins with a period (.)

Any command that begins with a slash mark (/)

➔ For more information, refer to [T5500 Series Tester Utility Program Reference Manual\(No. : 8256295\)](#).

- Commands accepted when the TSTG lamp on the control box is lit and test mode has been set to MANUAL:

CTRL/C command

CTRL/S command

CTRL/Q command

Space command

M command

F command

J command

V command

Z command

L command

N command

O command

H command

CTRL/B command

CTRL/D command

Any command that begins with a period (.)

Any command that begins with a slash mark (/)

S command

T command

P command

CTRL/T command

CTRL/I command

- Commands accepted when the CTRL/B command (key break) is being executed and the system is waiting for ↵ (CR) to be input:

CTRL/C command

CTRL/S command

CTRL/Q command

Space command

M command

F command

J command

V command

Z command

L command

N command

O command

H command

CTRL/B command

CTRL/D command

Any command that begins with a period (.)

Any command that begins with a slash mark (/)

S command

T command

P command

CTRL/T command

CTRL/I command

- Commands accepted before control is returned from the functional test to the system (control is not returned because the test mode of the pattern generator (PG) has been set to REPEAT):

CTRL/C command

CTRL/S command

CTRL/Q command

T command

P command

S command

CTRL/T command

CTRL/I command

 Tip

These commands are accepted only when the test mode of the control box is set to MANUAL.

2. Operation

This chapter explains controller command operations.

2. 1 Space Command

This command works the same way as the START switch on the control box.

- Starts test program execution.
- Restarts temporarily halted test programs.
- Returns control from a loop to the system when the test mode for the control box has been set to MANUAL or RECYCLE (HOLD can be either way) and the program is looping.

The table below shows which station is used for the space command.

Control box [DISABLE] key status		When neither station 1 nor 2 has executed the test program	When either station 1 or 2 is executing the test program or the program is temporarily halted
Station 1 control box	Station 2 control box		
ON	ON	Station 1	Station at which the program is being executed or is temporarily halted.
OFF	ON	Station 1	
ON	OFF	Station 2	
OFF	OFF	Station 1	

2. 2 CTRL/C Command

This command works the same way as the RESET switch on the control box. It forcibly aborts a running test program.

- Forcibly aborts a running key command.
- Clears the indication on the corresponding station.
- Clears all set tester conditions.

The station reset by the CTRL/C command is the same as the one reset by the space command.

The CTRL/C command and the RESET switch on the control box can be disabled by the utility program (/CKBRCNT).

➔ For more information on how to operate /CKBRCNT, refer to [T5500 Series Tester Utility Program Reference Manual\(No. : 8256295\)](#).

2. 3 CTRL/S, CTRL/Q, NO SCROLL (HOLD SCREEN) Commands

CTRL/S command

Up to 24 lines of strings can be displayed at a time on the VKT screen. When display continues to line 24, the first line is cleared from the screen and the second line becomes the top line. That is, the entire screen is scrolled upward one line. Therefore, use the CTRL/S command to stop scrolling when displaying more than 24 lines on the VKT screen.

Entering the CTRL/S key from the VKT temporarily halts the screen scroll and causes the ONLINE lamp (red) on VKT to blink. (Or the HOLD SCREEN lamp (red) will come on.)

CTRL/Q command

The CTRL/Q key command is used to restart scrolling temporarily halted by the CTRL/S command. The ONLINE lamp (red) on the VKT stops blinking and stays lit. (Or the HOLD SCREEN lamp will be turned off.) At the same time, the screen characters are scrolled up.

NO SCROLL (HOLD SCREEN) command

Each time the user presses this key, output to the VKT screen alternates between the CTRL/S and CTRL/Q codes.

2. 4 O Command

This command is used to select the mode when the test program name or manual analysis is specified.

This command enables setting control box functions included in each station from the VKT. The functions are as follows:

- Displays the test results generated up to this point.
Displays the name of the program last executed, test number, REMARKs, pass/fail, sample number, test status, and category lamp.
- Displays the currently-set program name, test mode, function key, data logging mode, start/stop test number, sample number, and DISABLE, INIT, and CAL.INIT settings.
- Determines whether or not to change the current specifications displayed and makes the required changes.

Below is an example of a specification by entering O:

When [O] is entered, the test results and current setting status are output as follows:

```

O      ← Nothing is displayed.

CONTROL BOX SELECTION  !
STATION NO.= 1        ← Enter the station number

PROGRAM NAME ...      FUNCTION TEST ...
TEST NO.    ... 0      PARAMETRIC TEST ...
REMARK      ...        SAMPLE NO.    ... 0
TEST STATUS ... READY

                                ↑ Displays the test results generated up to this point
                                ↓ Current setting status

CATEGORY ...
  MODE 1 : PRO NAME ...
  MODE 2 : TEST MODE ... AUTO
  MODE 3 : FK ...
  MODE 4 : DATA LOG ...
  MODE 5 : START TEST ... 0
  MODE 6 : STOP TEST ... 0
  MODE 7 : SAMPLE NO ... 0
  MODE 8 : INIT ...

SPECIFY MODE NUMBER THAT YOU WISH TO CHANGE.
MODE=
```

Enter a MODE number (1 to 8) to change the current setting.

If only ↵ (or a number other than 1 to 8) is entered at this time, the data settings will not be changed and command processing will be terminated.

When a MODE number (1 to 8) is entered, the system prompts you to change the current setting. Enter a new value.

The new values are output as shown below. (This screen is output regardless of which MODE has been changed.)

After confirming the change (output to the right of the corresponding MODE number), enter MODE=↓ (or any number other than 1 to 8) to terminate command processing. To make another change, enter MODE=1 to 8 (mode number of the item to be changed).

```
CONTROL BOX SELECTION  !
STATION NO.= 1

PROGRAM NAME ...      FUNCTION TEST ...
TEST NO.    ... 0      PARAMETRIC TEST ...
REMARK      ...      SAMPLE NO.    ... 0

TEST STATUS ... READY
CATEGORY

                                ↓   New settings
MODE 1 : PRO NAME ...
MODE 2 : TEST MODE ... AUTO
MODE 3 : FK ...
MODE 4 : DATA LOG ...
MODE 5 : START TEST ... 0
MODE 6 : STOP TEST ... 0
MODE 7 : SAMPLE NO ... 0
MODE 8 : INIT ...

SPECIFY MODE NUMBER THAT YOU WISH TO CHANGE.
MODE=
```

The following section explains the data specifications entered for each mode.

- When MODE = 1 is entered (to change the program name)

```
PREVIOUS PRO NAME =          ← Current specification
SPECIFY PRO NAME  = 8256↓    ← New specification
```

After the change, the display is as follows.

```
MODE 1 : PRO NAME ... 8256
```

- When MODE = 2 is entered (to change the test mode)

```
SPECIFY THE SWITCH, ENABLE(E) OR DISABLE(D)
MANUAL    RECYCLE    HOLD    FAIL-STOP FAIL-PAUS FAIL-INHI REPEAT DISABLE
  E         E         D         ↓
```

Enter **E** or **1** : ENABLE

Enter **D** or **0** : DISABLE

Enter **↵** : No change will be made hereafter

In the above example, MANUAL and RECYCLE are enabled and HOLD is disabled. After the change, the display is as follows.

```
MODE 2 : TEST MODE ... MANUAL RECYCLE
```

- When MODE = 3 is entered (to change the function key)

FK NO.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
PREVIOUS FK	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	← *1
SPECIFY FK	1	1	0	1	↵											

**1 Current settings*

Enter **E** or **1** : ON

Enter **D** or **0** : OFF

Enter **↵** : No change will be made hereafter

In the above example, FK 1, 2, and 4 set to ON.

After the change, the display is as follows.

```
MODE 3 : FK ... 1 2 4
```

- When MODE = 4 is entered (to change the data logging mode)

SPECIFY THE SWITCH, ENABLE (E) OR DISABLE (D)					
PASS	FAIL	FUNC	PARA	MARK	TEST-PIN TEST-DATA
E	E	E	E	↵	

Enter **E** or **1** : ENABLE

Enter **D** or **0** : DISABLE

Enter **↵** : No change will be made hereafter

In the above example, PASS, FAIL, FUNC, and PARA are enabled. After the data logging mode is changed, the specification for MODE 4 is output as follows:

```
MODE 4 : DATA LOG ... PASS-LOG FAIL-LOG FUNC-LOG PARA-LOG
MODE 5 : START TEST ... 0
MODE 6 : STOP TEST ... 0
```

- When MODE = 5, 6, and 7 are entered, display is as follows:

```
PREVIOUS START TEST NO.= 0
SPECIFY START TEST NO. =10↵

PREVIOUS STOP TEST NO. = 0
SPECIFY STOP TEST NO. =10↵

PREVIOUS SAMPLE NO. = 0
SPECIFY SAMPLE NO. =10↵
```

- When MODE = 8 is entered (to set INIT and CAL.INIT)

```
SPECIFY THE SWITCH, ENABLE (E) OR DISABLE (D)
INIT      CAL.INIT
E
```

Enter **E** or **1** : ENABLE

Enter **D** or **0** : DISABLE

Enter↵ : No change will be made hereafter

In the above example, INIT is changed to ENABLE. After the change, the display is as follows.

```
MODE 8 :  INIT      ... INIT
```

2. 5 S Command

This command selects the output device. Entering this command will display the following message:

```
OUTPUT DEVICE SELECTION!
OUTP (1-3) LPT(4) OUTP+FILE(5) FILE(6):DVN=_
```

If the TESTARIUM or ADVANstar-21 service computer is connected to the system, the following message is output:

```
OUTPUT DEVICE SELECTION!
OUTP(1-3) LPT(4) OUTP+FILE(5) FILE(6) OUTP+SERV(7) SERV(8): DVN=_
```

In response to this message, enter a device number (1 to 8) of the output device to be selected.

The output device numbers are as follows:

- 1 to 3 : System console (VKT)
- 4 : Line printer
- 5 : System console and specified file
- 6 : Specified files
- 7 : System console and specified service computer file
- 8 : Specified service computer file

If output device number 5, 6, 7, or 8 is entered, the following message is output:

```
SPECIFY LOG-FILE XXXX ____
```

In response to this message, enter the file name of the output file and press the [CR] key. When output device number 5 or 6 is specified, "customer name : file name" can also be specified.

In this case, the previously specified customer name and the file name are output. (The customer name is not output if it is the same as the current customer name.) If these names are not to going be changed, press only the [CR] key.

When the file name which has already existed is specified, the following message will be displayed.

```
____ HAS ALREADY EXISTED. SETUP HAS DONE !!
```

Note that the set up has already been done.

Tip

Note the following when specifying file names:

- File specifiers are not needed. ASC is assumed for all specifications other than file names that are unique to the system.
- The following file names are unique to the system:

File names unique to the satellite system:

\$LPT : Printer

MT0:n : Magnetic tape

\$CRO0 : VKT

\$CRO1 : VKT

File name unique to the service system:

\$LPT : Printer

The following data is output to the output device or file selected by this S command:

- Output data when the output device is specified by the WRITE statement as follows:
 WRITE ~ When no output unit is specified.
 WRITE [1, ~]~ When 1 is specified.
- Counter data to be listed out by the L command.
- Data to be output in the data logging mode.
- WRITE statement for the data logging mode; in other words, the output data when WRITE[n, ~]
 ~ and $101 \leq n \leq 131$ and $201 \leq n \leq 231$.
- Output data for the disk access check mode selected by the A command.
- Trace data selected by the CTRL/T command.

 Tip

- The system console (VKT) is the default output device when controller is started.
 - In the ADVANsite systems, a magnetic tape (MT0) cannot be selected as an output device and SERV cannot be specified as a service computer.
-

2. 6 CTRL/L Command

This command executes a form feed at the output device selected by the S command.

- When the output device is the system console (VKT): Clears the display screen.
- When the output device is an LPT: Outputs carriage return (CR) and form feed (FF) codes.

2. 7 I Command

This command is used to select an INPUT device. Entering this command will display the following message:

```
INPUT DEVICE SELECTION!
INP (1-3) PTR (4) FILE (5) :DVN=
```

In response to this message, enter a device number (1 to 5) of the output device to be selected. The input device numbers are as follows:

1 to 3 : System console (VKT)
5 : Disk file

When 5 is entered as the input device number, the following message is output:

```
SPECIFY INPUT-FILE XXXX __
```

In response to this message, enter a file name and press the [CR] key.

"customer name : file name" can also be specified.

In this case, XXXX is the previously specified customer name and the file name (The customer name is not printed if it is the same as the current customer name.) If these names are not going to be changed, press only the [CR] key.

Tip

Note the following when specifying file names:

- File specifiers are not needed. ASC is assumed for all specifications other than file names that are unique to the system.
- The following file names are unique to the system:

File names unique to the satellite system:

MT0:n : Magnetic tape

\$CRI0 : VKT

\$CRI1 : VKT

The input device selected by this I command is valid when the following INPUT statement is executed:

- INPUT~
- INPUT [n, ~] ~ (when n = 1)



Caution

- The system console (VKT) is the default input device when controller is started.
- In the ADVANsite systems, a magnetic tape (MT0) cannot be selected as an input device and SERV cannot be specified as a service computer.

2. 8 F Command

This command specifies the function key mode. Entering this command will display the following message:

```
FUNCTION-KEY MODE SELECTION!
```

```
FK1=0          <LF>
```

```
FK2=1          <LF>
```

```
FK3=5  8  9    <LF>
```

```
FK4=30         <LF>
```

```
FK6=0          <LF>
```

```
FK7=0          <LF>
```

```
FK8=0          <LF>
```

```
FK9=0          <LF>
```

```
FK10=0         <LF>
```

```
FK11=0         <LF>
```

```
FK12=0         <LF>
```

```
FK13=0         <LF>
```

```
FK14=0         <LF>
```

```
FK15=0         <LF>
```

```
FK16=0         <LF>
```

```
  ↑
```

```
*1
```

```
  ↑
```

```
*2
```

**1 Mode number previously set*

**2 Mode number to be newly set*

Enter mode numbers (0 to 32) to assign them to FK1 through FK16, respectively.

The user can specify two or more mode numbers for one FK by delimiting them by spaces or commas.

To move to the next FK, enter **<LF>** (line feed). To move to the next FK without making changes, enter only **<LF>**.

To terminate the mode number assignments, enter **↵** (carriage return) or press **[ESC]**.

Tip

The function key mode set here is valid when the function key with the corresponding MODE number is ON.

When two or more function keys are ON, all function key modes selected for these function keys are enabled.

Function key modes

Mode number	Meaning
1	Logs data used for the DC parametric test.
2	Logs failure data used for the DC parametric test.
1 and 2 together	Logs pass data used for the DC parametric test.
3	----
4	----
5	----
6	----
7	Uses AS1 to AS16 in place of FK1 to FK16.
8	Does not execute an entry operation of an INPUT statement. (Executes an output operation added to the INPUT statement.)
9	Does not execute an output operation of the statements added to WRITE, STOP, PAUSE, and INPUT statements.
10	Executes fail stop processing. (This mode is valid when AUTO has been specified. The MEAS statement is processed when HOLD is on, and the TEST statement is processed when HOLD is off.)
11	Executes fail pause processing. (This mode is valid when AUTO has been specified. The MEAS statement is processed when HOLD is on, and the TEST statement is processed when HOLD is off.)
12	----
13	----
14	Does not execute the program from the heading to the START TEST number. (This is equivalent to specifying GO TO TESTn at the head of a program.)
15	----
16	----
:	:
29	----
30	The INPUT statement setting is not displayed when it is entered.
31	Does not execute an output operation for the statement added to an INPUT statement.
32	Does not execute an output operation for the statements added to WRITE, STOP, or PAUSE statement.

2. 9 M Command

2. 9. 1 Outline of M Command

This command specifies data logging mode for the DC parametric test and functional test.

This command has the same functional ability as the /DTLOG utility program, but it uses conversation-type specification as opposed to the command-line type specification of /DTLOG.

The data logging mode is explained below.

- This mode is used to execute the DC parametric test, functional test, log all data, or log selected data while separating pass data from fail data.
- There are the following three data logging output formats for the DC parametric test:
 - LONG-type output format which outputs the test number, failure display, measurement results, LIMIT settings, or measured pin number in one line.
 - SHORT-type output format which outputs the test numbers, failure displays, and measured results for five tests in one line.
 - MARK log mode format which outputs the test numbers and failure displays for 12 tests in one line.
- For the T5593, T5592, T5591, T5586, T5585, T5581, T5581H, T5581P, T5571P, T5377, T5376, and T5375, a maximum of 64 failures can be logged. For the T5371, T5336, and T5311, a maximum of 16 failures can be logged. To analyze the process leading to the first failure, the failure itself and the preceding 15 patterns can be logged.
- Only arbitrary test entries can be data logged. This mode is called the test number data log mode.
- Data logging in the functional test can output the expected data every pin numbers of devices. This mode is called the pin number data log mode.
- Data logging in the functional test can output the pass-fail information of all devices.
- Data logging in the functional test can select the DUT.
If AFM MULTIMODE = SINGLE is specified by MEAS MPAT statement, the DUT number specified in AFM DUTSEL statement is assigned to be executed.

◆ Data Logging Specifications for Functional Tests

Target DUT for data logging

Data for a single specified DUT is logged.

The target DUT for data logging is the DUT specified with AFM DUTSEL=n.

Tip

*In addition to AFM DUTSEL=n, AFM MULTIMODE=SINGLE is also required.
AFM MULTIMODE=SINGLE is not required for the T5593.*

DUT selection (MODE8) data is valid only when MARK is not specified but FAIL and FUNC are specified in the data log mode.

Tip

When the DUT selection (MODE8) data is valid, the AFM MULTIMODE and AFM DUTSEL statements data are not recovered and therefore they are invalid.

ALPG operating mode for data logging

If data logging is executed for multiple fails (when the number of failures to be output is two or more in MODE3 or when the number of failures to be output is one and the process leading to the failure is not to be output), the ALPG operating mode is automatically set to the fail inhibit mode and the pattern program is executed to the end.

Pattern program in which both main PG and sub PG are used

Both main and sub fail data are logged.

Tip

*To execute a pattern program in which both main and sub patterns are written, AFM PGSEL=MAIN, SUB must be specified in the main program.
If not, incorrect address, timing set, MUT command, expected data, and fail data are displayed.*

DFM MODE statement

When data is logged, the DFM MODE setting is not recovered and therefore is invalid.

When data logging is not executed

- If the MARK log mode is set, functional test data is not logged.
 - Data is not logged under the conditions described below.
- When data is not logged, the messages shown below are displayed.

Tip

When the M command MODE7 is enabled, PASS/FAIL information on all devices is displayed.

- ◆ For the T5592, T5591, T5586, T5585, T5581, T5581H, T5581P, T5571P, T5377, T5376, T5375, T5371, T5336, and T5311, and when AFM MULTIMODE=ALL or CONFIGn is specified.

Data log can't work because AFM MULTIMODE is not SINGLE mode.

- ◆ For the T5586, T5585, T5581, T5581H, T5581P, T5571P, T5377, T5376, T5375, and T5371 and when AFM ACTION=PM>EXP is specified.

Data log can't work because AFM ACTION is only PM>EXP.

- ◆ For the T5377, T5376, T5375, and T5371, and when AFM ACTION=PM>DBM is specified.

Data log can't work because AFM ACTION is only PM>DBM.

- ◆ For the T5593 and when Parallel DFM is enabled.

Data log can't work because Parallel DFM is enable.

- When the ALPG is in the WAIT state, log data is not output.
- For the T5377, T5376, T5375, T5371, T5336, and T5311, fail log data is not displayed because the WSTRB3 and WSTRB4 failure signals cannot be stored in AFM and DFM.

◆ Expected Data in the Functional Test Data Log Display

Test program using ARIRAM conversion

Data before ARIRAM conversion is displayed as expected data.

However, when either of the following conditions is met in a program using ARIRAM conversion, data after ARIRAM conversion is displayed as expected data:

For the T5592, T5591, T5586, T5585, T5581H, T5581P, and T5571P

- D<TP2 and D<TPXOR cycles in a pattern program in which MODE TPM36 is specified but MODE TP2INV is not specified.
- D<TP2 and D<TPXOR cycles in a pattern program in which MODE TPM36 and MODE TP2INV are not specified and D0 through D17 are specified as expected data of the pin statement.
- D<BR and D<CBM cycles in a pattern program.

For the T5593

- MODE DBFORM4 is specified in a pattern program and SD0 through SD17 are specified as expected data of the pin statement.

"Z" display

For the T5593 and when the Hi-Z comparison is made, "Z" is displayed as expected data.

"?" display

In the following case, expected data is displayed as "?".

- ESM is specified in the pin statement.
- DSS2, DSS3 and DSS4 are specified in the pin statement.
- Data on and after pin 161 when AFM BITSIZE=n1, n2 (n1>36) and MODE TPM36 is specified.
- PPAT is specified in the pin statement.
- For the T5592, T5591, T5586, T5585, T5581, T5581H, T5581P, T5571P, T5377, T5376, T5375 and T5371 and when the AFM board is not installed in the system.

Tip

For the T5593, expected data is not displayed as "?".

When SDM, PSM, RDSM, RINV, DNRZ, and SMC are specified in the pin statement

Even when SDM, PSM, RDSM, RINV, DNRZ, and SMC are specified in the pin statement, expected data is displayed on the assumption that they are not specified.

For the T5593, expected data is displayed according to the setting.

When the DSC instruction is included in a pattern program

Regardless of whether or not the DSC instruction is included in a pattern program, expected data is displayed under the DSC=#1 condition.

For the T5593, expected data is displayed according to the specified DSC instruction.

◆ Fail Information in the Functional Test Data Log Display

Fail data logging

For the T5592, T5591, T5586, T5585, T5581, T5581H, T5581P, T5571P, T5377, T5376, T5375, and T5371, and when AFM ACTION is specified, fail data specified by the AFM ACTION statement is logged.

For the T5593, all fail data is logged regardless of the AFM ACTION setting.

Valid bit for fail information

For the T5592, T5591, T5586, T5585, T5581P, T5571P, T5377, T5376, T5375, and T5371

When $n_1 > n_2$ in the AFM BITSIZE= n_1 , n_2 statement, the number of n_1 bit is valid. (Data in n_2 is invalid.)

! Important

- The AFM BITSEL setting is invalid.
- For the T5371 and when MEAS MPAT(5) is used.

As shown below, the block indicated by "*" is the failure information for one of the tested devices. It does not indicate the information of the focused target DUT.

```
TEST 0100 FAIL DUT0001 MPAT PAT1
F      Y:#0010 X:#0021 TS:01 PC:#009 CYP:#01 CMD:R W
      1.1....1
      D:7.5....0....5....0
      EXP:000100001100001111
      FAIL:      #
```

} *

For the T5581 and T5581H

When a bit is selected (converted) by the AFM BITSEL statement, the fail information is displayed in the position of the selected (converted) bit.

When $n_1 > n_2$ in the AFM BITSIZE= n_1 , n_2 statement, the number of n_2 bits from the right edge of the displayed bit string is valid.

When n_1 is 18 bits or less by counting data from the rightmost bit
 n_2 bits from the first bit are valid.

When n_1 is 32 bits by counting data from the rightmost bit
 $n_2/2$ bits from the first bit and $n_2/2$ bits from the 17th bit are valid.

When n_1 is 36 bits by counting data from the rightmost bit
 $n_2/2$ bits from the first bit and $n_2/2$ bits from the 19th bit are valid.

When n_1 is 64 bits by counting data from the rightmost bit
 $n_2/4$ bits from the first bit, $n_2/4$ bits from the 17th bit, $n_2/4$ bits from the 33rd bit, and $n_2/4$ bits from the 49th bit are valid.

2. 9. 2 Operating Instructions for the M Command

The following message appears when the **M** command is entered:

```

M      ←The input value is not displayed when it is entered.

DATA LOG MODE SELECTION !
STATION NO.=2    ←Station number
                  ↓          Current setting status
MODE 1 : DATA LOG MODE      = PASS FAIL PARA
MODE 2 : DCT TYPE MODE       = LONG
MODE 3 : FCT TYPE MODE       = 1'ST FAIL ONLY.
MODE 4 : SAMPLE DEVICE LOG   = DISABLE
MODE 5 : TEST NO. LOG        = DISABLE
MODE 6 : PIN NO. MODE        = DISABLE
MODE 7 : ALL DUT GO/NOGO     = DISABLE
MODE 8 : SET DUT NO.         = DISABLE

SPECIFY MODE THAT YOU WISH TO CHANGE.
      MODE= 1
          ↑
      Enter a mode number to change the current setting.

```

When **1** is entered here, the following is displayed:

```

SPECIFY THE SWITCH, ENABLE(E) or DISABLE(D)
      PASS  FAIL  FUNC  PARA  MARK
        E    E    E    D    D

```

In response to this message, enter **E** or **D** to turn the required switch on or off.

Again, the menu is output as follows:

```

DATA LOG MODE SELECTION !
STATION NO.= 2

MODE 1 : DATA LOG MODE      = PASS FAIL FUNC
MODE 2 : DCT TYPE MODE       = LONG
MODE 3 : FCT TYPE MODE       = 1'ST FAIL ONLY.
MODE 4 : SAMPLE DEVICE LOG   = DISABLE
MODE 5 : TEST NO. LOG        = DISABLE
MODE 6 : PIN NO. MODE        = DISABLE
MODE 7 : ALL DUT GO/NOGO     = DISABLE
MODE 8 : SET DUT NO.         = DISABLE

SPECIFY MODE THAT YOU WISH TO CHANGE.
      MODE= 2
          ↑
      Enter a different mode number to change the current setting.

```

When **2** is entered here, the following is displayed:

```
SPECIFY DCT TYPE-MODE.  LONG(L) OR SHORT(S) ? S
```

For this message, enter the output type of the DC parametric test data logging: Again, the menu is output as follows:

```
DATA LOG MODE SELECTION !
STATION NO.= 2

MODE 1 : DATA LOG MODE      = PASS FAIL FUNC
MODE 2 : DCT TYPE MODE      = SHORT
MODE 3 : FCT TYPE MODE      = 1'ST FAIL ONLY.
MODE 4 : SAMPLE DEVICE LOG  = DISABLE
MODE 5 : TEST NO. LOG       = DISABLE
MODE 6 : PIN NO. MODE       = DISABLE
MODE 7 : ALL DUT GO/NOGO    = DISABLE
MODE 8 : SET DUT NO.        = DISABLE

SPECIFY MODE THAT YOU WISH TO CHANGE.
      MODE= 3
            ↑
      Enter a different mode number to change the current setting.
```

If **3** is entered here, the system asks for the number of failures to be output for data logging in the functional test.

```
SPECIFY FCT TYPE-MODE
SPECIFY FAIL COUNT = 1↓
```

Enter the number of failures to be output. For the T5593, T5592, T5591, T5585, T5581, T5581H, T5581P, T5571P, T5377, T5376, and T5375, a maximum of 64 failures can be entered. For the T5371, T5336, and T5311, a maximum of 16 failures can be entered.

If 1 is entered for the number of failures, the system prompts whether or not the process leading to the failure should also be output.

```
DO YOU WISH TO SHOW THE CAREER, YES(Y) OR NO(N) ? N
```

Enter **Y** or **N**.

Again, the menu is output as follows:

```
DATA LOG MODE SELECTION !
STATION NO.= 2

MODE 1 : DATA LOG MODE      = PASS FAIL FUNC
MODE 2 : DCT TYPE MODE       = SHORT
MODE 3 : FCT TYPE MODE       = 1'ST FAIL ONLY.
MODE 4 : SAMPLE DEVICE LOG   = DISABLE
MODE 5 : TEST NO. LOG        = DISABLE
MODE 6 : PIN NO. MODE        = DISABLE
MODE 7 : ALL DUT GO/NOGO     = DISABLE
MODE 8 : SET DUT NO.         = DISABLE

SPECIFY MODE THAT YOU WISH TO CHANGE.
MODE= 5
  ↑
Enter a different mode number to change the current setting.
```

If **5** is entered here, the system asks whether or not the test number data logging is to be carried out.
If you do, enter test numbers.

```
SPECIFY TEST NO. DATA LOG MODE. ENABLE(E) OR DISABLE(D) ? E
SPECIFY START TEST NO. & STOP TEST NO.
START-STOP TEST NO.= 1-10<LF>    ← When <LF>, prompts for the next input
START-STOP TEST NO.= 20-30<LF>
START-STOP TEST NO.= 50<LF>
START-STOP TEST NO.= 60J        ← Terminate the program with J
```

Again, the menu is output as follows:

```
DATA LOG MODE SELECTION !
STATION NO.= 2

MODE 1 : DATA LOG MODE      = PASS FAIL FUNC
MODE 2 : DCT TYPE MODE       = SHORT
MODE 3 : FCT TYPE MODE       = 1'ST FAIL ONLY.
MODE 4 : SAMPLE DEVICE LOG   = DISABLE
MODE 5 : TEST NO. LOG        = ENABLE
      START-STOP TEST        = 1-10 2-30 50 60
MODE 6 : PIN NO. MODE        = DISABLE
MODE 7 : ALL DUT GO/NOGO     = DISABLE
MODE 8 : SET DUT NO.         = DISABLE

SPECIFY MODE THAT YOU WISH TO CHANGE.
MODE= 6
  ↑
Enter a different mode number to change the current setting.
```

When **6** is entered here, the following is displayed:

```
OUT PUT PIN NO.MODE.  ENABLE(E) OR DISABLE(D) ?E
```

For this message, enter the output type of the functional test data logging:
The following is displayed again.

```
DATA LOG MODE SELECTION !
STATION NO.= 2

MODE 1 : DATA LOG MODE      = PASS FAIL FUNC
MODE 2 : DCT TYPE MODE       = SHORT
MODE 3 : FCT TYPE MODE       = 1'ST FAIL ONLY.
MODE 4 : SAMPLE DEVICE LOG   = DISABLE
MODE 5 : TEST NO. LOG        = ENABLE
      START-STOP TEST        = 1-10 2-30 50 60
MODE 6 : PIN NO. MODE        = ENABLE
MODE 7 : ALL DUT GO/NOGO     = DISABLE
MODE 8 : SET DUT NO.         = DISABLE

SPECIFY MODE THAT YOU WISH TO CHANGE.
      MODE= 7
            ↑
      Enter a different mode number to change the current setting.
```

When **7** is entered here, the following is displayed:

```
OUT PUT ALL DUT GO/NOGO DATA ENABLE(E) OR DISABLE(D) ? E
```

For this message, enter the output type of the functional test data logging:

The following is displayed again.

```
DATA LOG MODE SELECTION !
STATION NO.= 2

MODE 1 : DATA LOG MODE      = PASS FAIL FUNC
MODE 2 : DCT TYPE MODE       = SHORT
MODE 3 : FCT TYPE MODE       = 1'ST FAIL ONLY.
MODE 4 : SAMPLE DEVICE LOG   = DISABLE
MODE 5 : TEST NO. LOG        = ENABLE
      START-STOP TEST        = 1-10 2-30 50 60
MODE 6 : PIN NO. MODE        = ENABLE
MODE 7 : ALL DUT GO/NOGO     = ENABLE
MODE 8 : SET DUT NO.         = DISABLE

SPECIFY MODE THAT YOU WISH TO CHANGE.
MODE= 8
      ↑
      Enter a different mode number to change the current setting.
```

When **8** is entered here, the following is displayed:

```
SET DUT NO.      ENABLE (E) OR DISABLE (D) ? E
```

In response to this display, enter whether or not the DUT number is changed for data logging in a functional test.

If **E** is entered here, the system prompts to enter the DUT number to be changed.

For the T5592, T5591, T5586, T5585, T5581, T5581H, T5581P, T5571P, T5377, T5376, T5375, T5371, T5336, and 5311

```
AFM MULTIMODE = SINGLE
AFM DUTSEL      = 5
```

The AFM MULTIMODE statement is changed to SINGLE mode.

For the T5593

```
SET DUT NO. = 5
```

Enter the DUT number to be changed.

The following is displayed again.

```
DATA LOG MODE SELECTION !
STATION NO.= 2

MODE 1 : DATA LOG MODE      = PASS FAIL FUNC
MODE 2 : DCT TYPE MODE       = SHORT
MODE 3 : FCT TYPE MODE       = 1'ST FAIL ONLY.
MODE 4 : SAMPLE DEVICE LOG   = DISABLE
MODE 5 : TEST NO. LOG        = ENABLE
      START-STOP TEST        = 1-10 2-30 50 60
MODE 6 : PIN NO. MODE        = ENABLE
MODE 7 : ALL DUT GO/NOGO     = ENABLE
MODE 8 : SET DUT NO.         = 5  (AFM MULTIMODE=SINGLE)
                                ↑
SPECIFY MODE THAT YOU WISH TO CHANGE.      Not displayed for the T5593.
MODE= 1
```

If a mode number other than 1 through 8 is entered here, this M command is terminated.

! Important

The MODE 4 (SAMPLE DEVICE LOG) cannot be used with ASX/U-51.

2. 9. 3 Expected Data and Fail Information of Functional Tests

Expected data and fail information of a functional test are displayed as shown below. The data varies depending on the system.

For the T5581, T5581H, T5581P, and T5571P

- MAIN

EXP:0001 Expected data (Odd phase STRB)

FAIL: # Fail information

- SUB

EXP:0010 Expected data (Even phase STRB)

FAIL: # Fail information

For the T5591

- The MAIN PG when the AFM double-speed mode is not included in the AFM ADDRESS statement

EXP001:0001 Expected data (Odd phase STRB from STRB1 to 128)

EXP129:0100 Expected data (Odd phase STRB from STRB129 to 256)

FAIL: # # Fail information

- The SUB PG when the AFM double-speed mode is not included in the AFM ADDRESS statement

EXP002:0010 Expected data (Odd phase STRB from STRB1 to 128)
 EXP130:1000 Expected data (Odd phase STRB from STRB129 to 256)
 FAIL:# # Fail information

- When the AFM double-speed mode is included in the AFM ADDRESS statement

EXP001:0001 Expected data (Odd phase STRB from STRB1 to 128)
 FAIL: # Fail information
 EXP129:0100 Expected data (Odd phase STRB from STRB129 to 256)
 FAIL: # Fail information

For the T5336 and T5311

EXP:0001 Expected data (Odd phase STRB)
 READ FUNC: # Fail information

For the T5377, T5376, T5375, and T5371

- When the AFM double-speed mode is not included in the AFM ADDRESS statement

EXP:0001 Expected data (Odd phase STRB)
 FAIL: # Fail information

- When the AFM double-speed mode is included in the AFM ADDRESS statement

EXP001:0001 Expected data (Odd phase STRB)
 FAIL: # Fail information
 EXP002:0010 Expected data (Even phase STRB)
 FAIL: # Fail information

- When the AFM double-speed mode (2 paths) is included in the AFM ADDRESS statement
 When SB1 is specified for the strobe in the AFM ACTION statement

EXP001:0001 Expected data (Odd phase STRB)
 FAIL: # Fail information

When SB2 is specified for the strobe in the AFM ACTION statement

EXP002:0010 Expected data (Even phase STRB)
 FAIL: # Fail information

For the T5586 and T5585

- AFM PGSEL=MAIN when the AFM double-speed mode is not included in the AFM ADDRESS statement

EXP001:0001 Expected data (Odd phase STRB from STRB1 to 128)
 EXP129:0100 Expected data (Odd phase STRB from STRB129 to 256)
 FAIL: # # Fail information

- When the AFM double-speed mode is included in the AFM ADDRESS statement

EXP001:0001 Expected data (Odd phase STRB from STRB1 to 128)
 FAIL: # Fail information
 EXP129:0010 Expected data (Odd phase STRB from STRB129 to 256)
 FAIL: # Fail information

- When the AFM double-speed mode (2 paths) is included in the AFM ADDRESS statement

When SB1 is specified for the strobe in the AFM ACTION statement

EXP001:0001 Expected data (Odd phase STRB from STRB1 to 128)
 FAIL: # Fail information

When SB129 is specified for the strobe in the AFM ACTION statement

EXP129:0010 Expected data (Odd phase STRB from STRB129 to 256)
 FAIL: # Fail information

- AFM PGSEL=MAIN, SUB when the AFM double-speed mode is not in the AFM ADDRESS statement

MAIN

EXP001:0001 Expected data (Odd phase STRB from STRB1 to 128)
 FAIL: # Fail information

SUB

EXP002:0010 Expected data (Even phase STRB from STRB1 to 128)
 FAIL: # Fail information

For the T5592

When PPAT is specified in the PIN statement, DBM pattern data (consisting of 0, 1, L, H, and X) is displayed in the DBM001 and DBM129. However, the DBM pattern data is not displayed for a cycle in which INHDBM is specified in the pattern program.

- When the AFM double-speed mode is not in the AFM ADDRESS statement
 For a cycle in which INHDBM is not specified in the pattern program and PPAT is specified in the PIN statement

DBM001:LLLH DBM pattern information (Odd phase STRB from STRB1 to 128)
 EXP001:0001 Expected data (Odd phase STRB from STRB1 to 128)
 DBM129:LHLL DBM pattern information (Odd phase STRB from STRB129 to 256)
 EXP129:0100 Expected data (Odd phase STRB from STRB129 to 256)
 FAIL: # # Fail information

For a cycle in which INHDBM is specified in the pattern program and PPAT is specified in the PIN statement

DBM001: DBM pattern information (Odd phase STRB from STRB1 to 128)
 EXP001:0001 Expected data (Odd phase STRB from STRB1 to 128)
 DBM129: DBM pattern information (Odd phase STRB from STRB129 to 256)
 EXP129:0100 Expected data (Odd phase STRB from STRB129 to 256)
 FAIL: # # Fail information

When PPAT is not specified in the PIN statement

DBM001: DBM pattern information (Odd phase STRB from STRB1 to 128)
 EXP001:0001 Expected data (Odd phase STRB from STRB1 to 128)
 DBM129: DBM pattern information (Odd phase STRB from STRB129 to 256)
 EXP129:0100 Expected data (Odd phase STRB from STRB129 to 256)
 FAIL: # # Fail information

The following are displayed when PPAT is specified in the PIN statement.

- When the AFM double-speed mode is in the AFM ADDRESS statement

DBM001:LLLH DBM pattern information (Odd phase STRB from STRB1 to 128)

EXP001:0001 Expected data (Odd phase STRB from STRB1 to 128)

FAIL: # Fail information

DBM129:LHLL DBM pattern information (Odd phase STRB from STRB129 to 256)

EXP129:0100 Expected data (Odd phase STRB from STRB129 to 256)

FAIL: # Fail information

- When the AFM double-speed mode (2 paths) is in the AFM ADDRESS statement

When SB1 is specified for the strobe in the AFM ACTION statement

DBM001:LLLH DBM pattern information (Odd phase STRB from STRB1 to 128)

EXP001:0001 Expected data (Odd phase STRB from STRB1 to 128)

FAIL: # Fail information

When SB129 is specified for the strobe in the AFM ACTION statement

DBM129:LHLL DBM pattern information (Odd phase STRB from STRB129 to 256)

EXP129:0100 Expected data (Odd phase STRB from STRB129 to 256)

FAIL: # Fail information

For the T5593

- AFM PGSEL=MAIN when the AFM double-speed mode is not included in the AFM ADDRESS statement

EXP001:0001 Expected data (Odd phase STRB from STRB1 to 128)

EXP129:0100 Expected data (Odd phase STRB from STRB129 to 256)

FAIL: # # Fail information

- AFM PGSEL=MAIN when the AFM double-speed mode is included in the AFM ADDRESS statement

EXP001:0001 Expected data (Odd phase STRB from STRB1 to 128)

FAIL: # Fail information

EXP129:0100 Expected data (Odd phase STRB from STRB129 to 256)

FAIL: # Fail information

- AFM PGSEL=MAIN when the AFM double-speed mode (2 paths) is included in the AFM ADDRESS statement

When SB1 is specified for the strobe in the AFM ACTION statement

EXP001:0001 Expected data (Odd phase STRB from STRB1 to 128)

FAIL: # Fail information

When SB129 is specified for the strobe in the AFM ACTION statement

EXP129:0010 Expected data (Odd phase STRB from STRB129 to 256)

FAIL: # Fail information

- AFM PGSEL=MAIN, SUB when the AFM double-speed mode is not included in the AFM ADDRESS statement

MAIN

EXP001:0001 Expected data (Odd phase STRB from STRB1 to 128)

EXP129:0100 Expected data (Odd phase STRB from STRB129 to 256)

FAIL: # # Fail information

SUB

EXP002:0010 Expected data (Even phase STRB from STRB1 to 128)

EXP130:1000 Expected data (Even phase STRB from STRB129 to 256)

FAIL: # # Fail information

- AFM PGSEL=MAIN, SUB when the AFM double-speed mode is included in the AFM ADDRESS statement

MAIN

EXP001:0001 Expected data (Odd phase STRB from STRB1 to 128)

FAIL: # Fail information

EXP129:0100 Expected data (Odd phase STRB from STRB129 to 256)

FAIL: # Fail information

SUB

EXP002:0010 Expected data (Even phase STRB from STRB1 to 128)

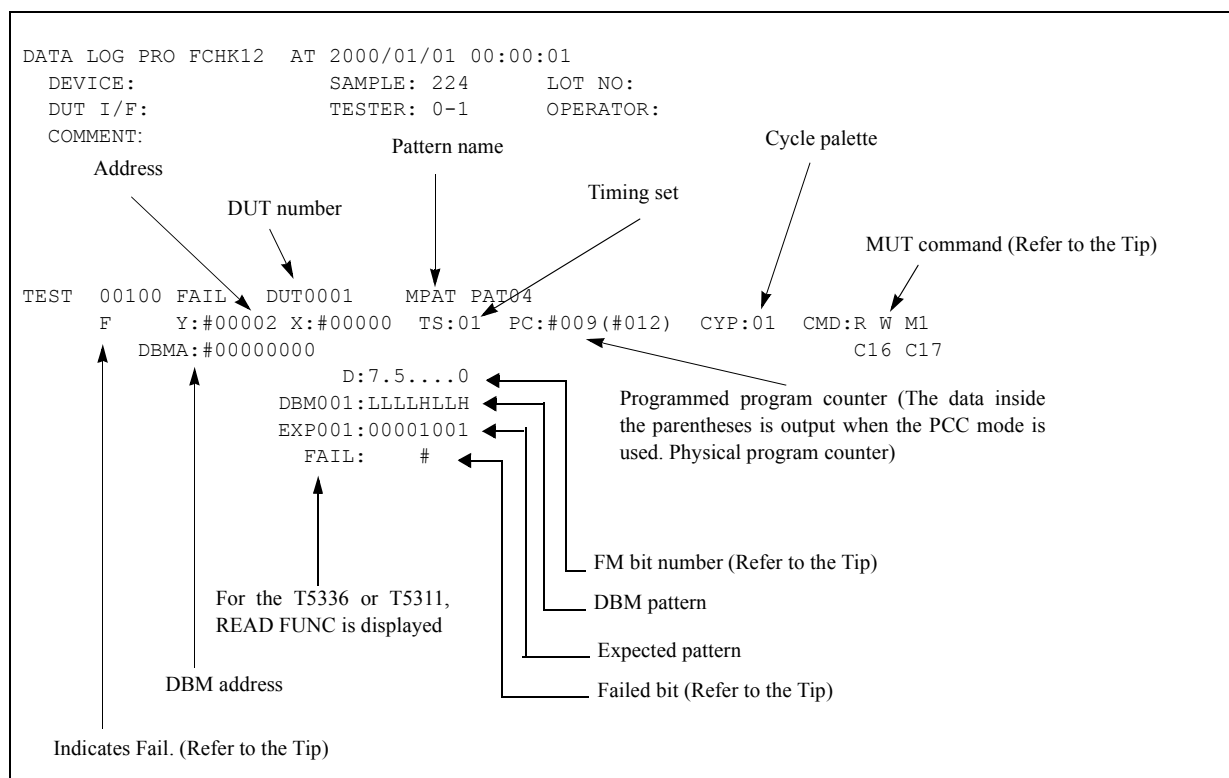
FAIL: # Fail information

EXP130:1000 Expected data (Even phase STRB from STRB129 to 256)

FAIL: # Fail information

2. 9. 4 Functional Test Output Example

Output example of first failure



Tip

- The FM bit number is expressed using the multiples of 0 and 5 and a maximum bit number, which are delimited by a period (.).
- Character F showing a fail occurred is changed to ?, depending on the fail notation.
- The fail information for the T5336 and T5311 is the data which was read out by READ FUNC.
- The MUT commands C16 to C23 are displayed on the next line by the line feed.
- For the T5593 and T5377, when the AFM double-speed mode is included in the AFM ADDRESS statement, the Y addresses are displayed in two lines as Y-1ST (the first Y address) and Y-2ND (the second Y address).

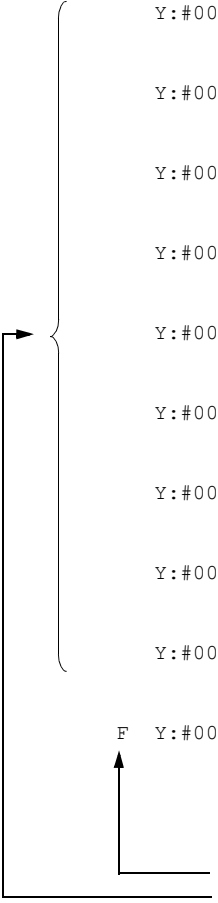
Output example including the process to the failure

```

DATA LOG PRO FCHK12   AT 1999/06/20 16:31:47
DEVICE:                SAMPLE: 233      LOT NO:
DUT I/F:               TESTER: 0-1      OPERATOR:
COMMENT:

TEST 00100   FAIL   DUT0001  MPAT PAT04
  Y:#0003 X:#0003  TS:01  PC:#001      CMD:  W M1
    D: 7.5....0
    EXP: 00000000
  Y:#0000 X:#0000  TS:01  PC:#001      CMD:  W M1
    D: 7.5....0
    EXP: 00000001
  Y:#0000 X:#0001  TS:01  PC:#001      CMD:  W M1
    D: 7.5....0
    EXP: 00000010
  Y:#0000 X:#0002  TS:01  PC:#001      CMD:  W M1
    D: 7.5....0
    EXP: 10000011
  Y:#0000 X:#0003  TS:01  PC:#001      CMD:  W M1
    D: 7.5....0
    EXP: 00000100
  Y:#0001 X:#0000  TS:01  PC:#001      CMD:  W M1
    D: 7.5....0
    EXP: 00000101
  Y:#0001 X:#0001  TS:01  PC:#001      CMD:  W M1
    D: 7.5....0
    EXP: 00000110
  Y:#0001 X:#0002  TS:01  PC:#001      CMD:  W M1
    D: 7.5....0
    EXP: 00000111
  Y:#0001 X:#0003  TS:01  PC:#001      CMD:  W M1
    D: 7.5....0
    EXP: 10001000
F Y:#0002 X:#0000  TS:01  PC:#001      CMD:R W M1
  D: 7.5....0
  EXP: 00001001
  FAIL:      #

```


(a)

Failure location
Patterns to the failure

 Tip

The data log details (a) will be incorrect if a failure occurs only for a pin listed below because the failure is unable to be stored in the DFM.

T5592:

Pin 51 to 56, 115 to 120, 179 to 184, or 243 to 248

T5591:

Pin 42, 43, 58, 59, 106, 107, 122, 123, 170, 171, 186, 187, 234, 235, 250, 251, 298, 299, 314, 315, 362, 363, 378, 379, 426, 427, 442, 443, 490, 491, 506, or 507

T5377, T5376, T5375, and T5371:

Pin 51, 52, 115, or 116

Pin 49, 50, 113, or 114 when PIN MODE=C for the socket program

Output example of pass failure information and pin number of all devices

```
DATA LOG  PRO FCHK12  AT 1999/06/20 17:07:14
DEVICE:          SAMPLE: 100      LOT NO:
DUT I/F:         TESTER: 0-1      OPERATOR:
COMMENT:

TEST 00100 FAIL  DUT0001  MPAT PAT07  <P:PASS  *:FAIL  -:NO TESTING >
TEST DUT       : .....v
< 0001-0008 >  **P*PPPP
  ↑
  The number of simultaneous measurements

F      Y:#0002 X:#0000  TS:01  PC:#009      CMD:R W M1
                        43333333
                        PC: 09876543  ) ← Pin number
                        EXP: 00001001
                        FAIL:      #
```



Tip

Mark v is output every eight dots.

Output example of specifying AFM PGSEL=MAIN, SUB

```

DATA LOG  PRO FCHK12   AT 1999/06/20 16:39:59
DEVICE:                SAMPLE: 110      LOT NO:
DUT I/F:               TESTER: 0-1      OPERATOR:
COMMENT:

TEST 00100 FAIL  DUT0001  MPAT PAT07
MAIN F      Y:#0002 X:#0002  TS:01  PC:#009      CMD:R W      C0
              1.1....1....1
              D: 2.0....5....0....5....0
              EXP: 00000111001010000100111
              FAIL:      ### # #      # ###
MAIN F      Y:#0004 X:#0004  TS:01  PC:#00B      CMD:R W      C1
              1.1....1....1
              D: 2.0....5....0....5....0
              EXP: 00000111001010000101001
              FAIL:      ### # #      # # #
SUB  F      Y:#0002 X:#0002  TS:01  PC:#009      CMD:R W      C2
              1.1....1....1
              D: 2.0....5....0....5....0
              EXP: 11111000110101111011000
              FAIL: #####  ## # ##### ##
SUB  F      Y:#0004 X:#0004  TS:01  PC:#00B      CMD:R W      C3
              1.1....1....1
              D: 2.0....5....0....5....0
              EXP: 11111000110101111010110
              FAIL: #####  ## # ##### # ##

```



Tip

MAIN or SUB is output only when AFM PGSEL=MAIN,SUB was specified.

Example of MARK log mode data logging output for the DC parametric test

```
DATA LOG  PRO SAMPLE  AT 1999/06/20 14:12:35
DEVICE:  8256          SAMPLE: 1          LOT NO:  LT830404-12
DUT I/F: M6741A      TESTER: 0-1        OPERATOR:
COMMENT: FINAL TEST

                Failure display
                ↙
0001  0002 U0003 U0004 U0005  0006  0007  0008  0009  0010  0011  0012
0013  0014 L0015  0016 L0017  0018  0019  0020  0021  0022
↑
Test number
```

2.10 Z Command

This command displays and specifies ACTION SELECTOR.

Press the [Z] key from VKT.

```

← Press the [Z] key
ACTION SELECTOR!
SWITCH(0) OR MODE(1) OR BOTH(2) ?  0, 1 or 2
STATION?  1 or 2
BIT      0000000001111111
          1234567890123456
ON/OFF   0000000000000000
Input change bit data 1/0

AS01= Mode number<LF>
AS02=      :
AS03=      :
      :
      :
AS016= Mode number


```

Diagram annotations:

- SWITCH(0) OR MODE(1) OR BOTH(2) ? 0, 1 or 2: Specifies SWITCH (0)
- BIT 0000000001111111: Displays current bit status
- ON/OFF 0000000000000000: Displays current bit status
- AS01= Mode number<LF>: Specifies BOTH (2)
- AS02= : Specifies MODE (1)
- AS03= : Specifies MODE (1)
- : Specifies MODE (1)
- : Specifies MODE (1)
- AS016= Mode number: Specifies BOTH (2)

The Z command settings work in a similar way as the F command of FUNCTION KEY. It can specify the mode for the ACTION SELECTOR.

For example, to control MODE4 (prohibits execution of the WRITE! statement) using ACTION SELECTOR BIT 3, follow the setting procedure below to use ACTION SELECTOR BIT3 (AS03) as MODE4.

```

Z      ← Nothing is displayed.
ACTION SELECTOR!
SWITCH(0) OR MODE(1) OR BOTH(2) ? 2
STATION? 1
BIT      0000000001111111
          1234567890123456
ON/OFF   0000000000000000
001
AS01= 00 <LF>
AS02= 00 <LF>
AS03= 00 4<LF>

```

By entering the parts above, MODE4 is selected using the ACTION SELECTOR.

To scan only individual bit correspondence, enter 0 at SWITCH(0).

To scan correspondence of mode numbers, enter 1 at MODE(1).

To check to see the program status after entering the above, for example, enter **2** at BOTH (2).

```

Z      ← Nothing is displayed.
ACTION SELECTOR!
SWITCH(0) OR MODE(1) OR BOTH(2) ? 2
STATION? 1
BIT      0000000001111111
          1234567890123456
ON/OFF   0010000000000000
          ↵
AS01= 00  <LF>
AS02= 00  <LF>
AS03= 04  ↵

```

Action selector modes

Mode number	Meaning
0	Nothing is selected
1	Data file service mode
2	Executable program service mode
3	Mode to inhibit all functions for data files
4	Mode to inhibit only the WRITE! statement
5	Mode to inhibit only the READ! statement
6	Mode to generate a data log file
7	Data logging dual mode
8	Mode to inhibit output sample numbers to a data file
:	:
31	----

MODE1

Data file service mode

This mode specifies whether the data file is to be created as a disk file on the service computer side. If this mode is specified, the data file is created at the service computer. If this mode is not specified, the data file is created on the satellite tester side. (This specification is determined at execution of the OPEN! statement.)

MODE2

Executable program service mode

This mode specifies whether the program to be executed by the satellite tester is to be loaded to the system from a disk file on the service computer side or from a program file managed by the satellite tester.

If this mode is specified, the executable program will be loaded to the system from a disk file. If this mode is not specified, the executable program will be loaded to the system from a program file in the satellite tester.

Executable programs are the subprograms invoked by the CALL, MEAS, and SEND commands as well as the main program.



Caution

Service mode cannot be specified in the ADVANsite.

If MODE1 or MODE2 is selected by a satellite tester that is not connected to the service computer, the system assumes that this mode has not been specified and automatically processes a file on the satellite tester.

MODE3

Mode to inhibit all functions for data files

This mode inhibits functions associated with part of data file without changing the test program. Selection of this mode specifically inhibits the functions to read and write the data file, regardless of the service computer or satellite tester specifications.

MODE4

Mode to inhibit only the WRITE! statement relating to data files

This mode partially inhibits data file create (write) processing without changing the test program. This mode does not inhibit the REWRITE! statement. Also, the use of a certain statements in the program can partially inhibit write processing. The SET/RESET ASMODE statement must be used in this case.



Tip

Note that the functions other than that of the WRITE! statement, such as those of the OPEN! and CLOSE! statements, are active.

MODE5

Mode to inhibit only the READ! statement relating to data files

This mode inhibits the function of the READ! statement when data that has been read to the data file need not be changed. Therefore, the READ! statement is not executed and the previous data remains in a variable as it was.



Tip

The functions other than that of the READ! statement are still active.



Caution

For MODE3, MODE4, or MODE5, all functions are inactive when the open function (open for temporary operation) has an upward arrow sign, such as the OPEN!n↑statement.

This means the MODE3, MODE4, and MODE5 functions are ignored.

MODE6

Mode to generate a data log file

This mode is used to file data log information as is to a device specified as the data file when executing, at a satellite tester, a program that includes the DC parametric test and is capable of data logging. The data log information used for data filing is as follows:

1. Pass/fail information
2. UPPER/LOWER when FAIL occurs
3. Test number
4. Measurement data (DC parametric test)

These data logs 1 to 4 are packed into the following six-word format data files. These can be divided into data 1 to 3 (logical type) and data 4, general information data (current, voltage, etc.), to process.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	Variable									0	0	Logical-type code			
P/F	U/L														
Test number															
Packing information of 1,2, and 3															
0	Variable									ovr	0	Code			
General information data															

Measurement data of 4

- P/F: 0 for PASS, 1 for FAIL
- U/L: When FAIL occurs, 0 for UPPER and 1 for LOWER
(Always 0 for PASS)

A data file must be opened to create a data log file. Create a data log file using the file name specified by the J command.

To specify a channel for the data file, after entering a channel number from 1 to 32, the [D] key must be pressed for data logging.

```

J ←Press the [J] key
OPEN DATA FILE!
CHANNEL? nD↵
MODE? mode [[*]m]↵
FILENAME? filename↵
STATION? ns↵ or ↵
DEVICE? Device name↵
CONDITION Measurement conditions↵
LOT? Lot name-Sub-lot number↵
OPEN OK!

```

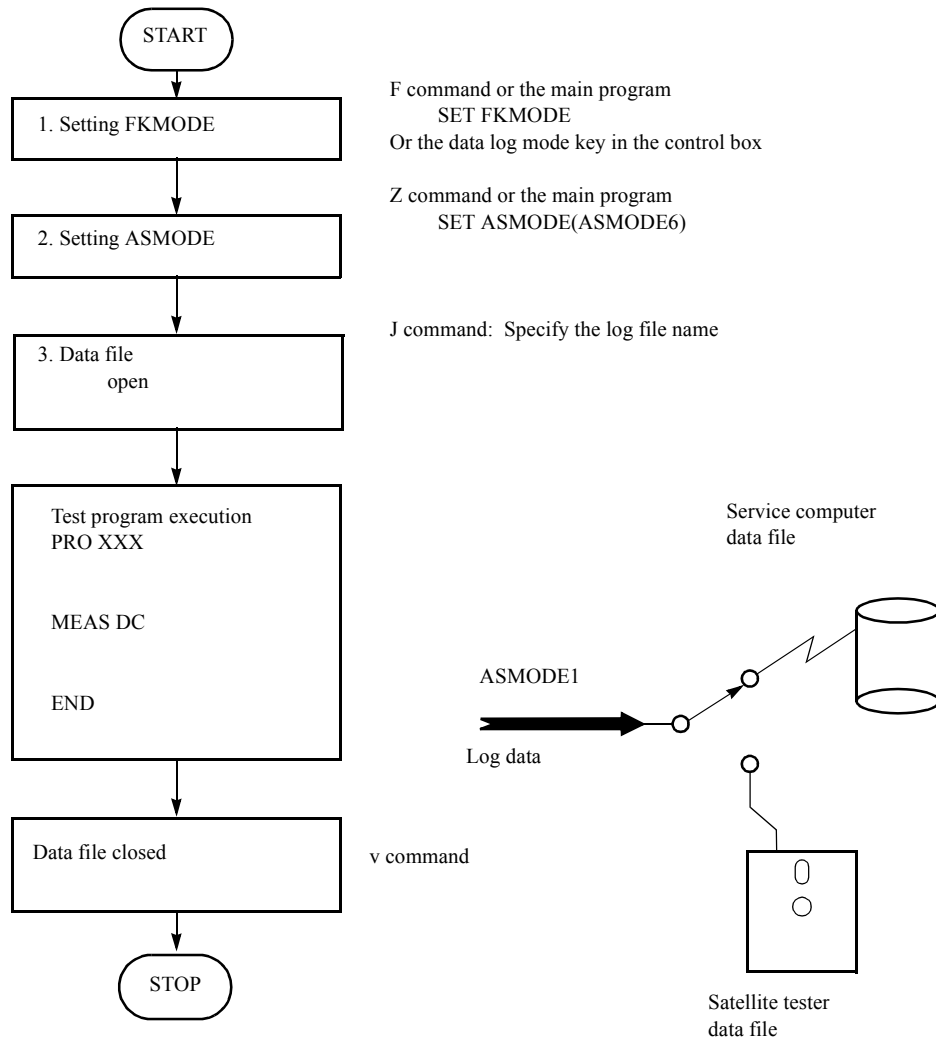
When opening a data log file, press [W] to select the mode. The [R] key is not allowed for opening data files. Use the key only when opening write files.

Conditions for data log file creation

To create a data log file, the following three conditions must be satisfied.

1. There are two ways to specify the data log mode:
 - Using the FUNCTION KEY MODE
 - MODE1 : ALL DATA LOG
 - MODE2 : FAIL DATA LOG
 - MODE1 and MODE2 : PASS DATA LOG
 - Using the data log mode key on the control box
 - When the MARK key is down, the functional test is omitted.
2. Set the ACTION SELECTOR MODE to the data log file creation mode.
 - MODE6 or MODE7
3. Use the J command to open the data log file.

The data log operation flow is as follows:



2.11 J Command

This command opens data files.

Pressing the [J] key from VKT will display the following message:

```

OPEN DATA FILE!
CHANNEL?  1↵      ..... (a)
MODE?    W↵      ..... (b)
FILENAME? ABC↵    ..... (c)
STATION?  1↵      ..... (d)
DEVICE?  SAMPLE↵ ..... (e)
CONDITION? STANDARD↵ ..... (f)
LOT?    A0001↵    ..... (g)

```

OPEN OK!

(a) Specify the channel number to open for creating a data file

If D is suffixed to the channel number, a data log file is created. The action selector specification determines whether log data is actually written to the file.

(b) Specify the data file mode

To create a write file, press the [W] key, and for a read file, press the [R] key. Specify using the W key when creating a data log file.

In this case, each data length requires 12 bytes.

If the mode is W, the file size can be specified. In this case, the file is to be a contiguous file.

W m : The file size is a multiple of 24 Kilobytes times m.

W *m: The file size is 2*m bytes.

Selecting %W sets the sample number write mode.

➔ For more information, refer to the section (setting sample number) in [2. 13 ".S Command" on page 2-46](#).

(c) Specify the data file name

Enter the file name of the data file to be created or the data file to be read.

(d) Specify the station number

If the header information is not required, simply enter ↵ (CR code) or N↵.

Header information will be required when the service computer starts processing data.

If ↵ is entered, a 4-byte keyword indicating that there is no header information is written at the head of the file. If N↵ is entered, nothing is written into the file.

(e) Specify the device name

Enter the device name.

This character string can be up to 12 characters long.

(f) Specify the character string information

Enter the character string information relating to measurement conditions.

This character string can be up to 12 characters long.

(g) Specify the lot name

Specify the information relating to the lot.

A lot name can be up to 12 characters long. Entering the - code following the lot name can specify a sub-lot number.

(Example)

TLCX174BB-72
└───┬───┘ └──┘
Lot name Sub-lot name

If the above information is entered correctly, the data file will be opened and the message "OPEN OK!" will be displayed.

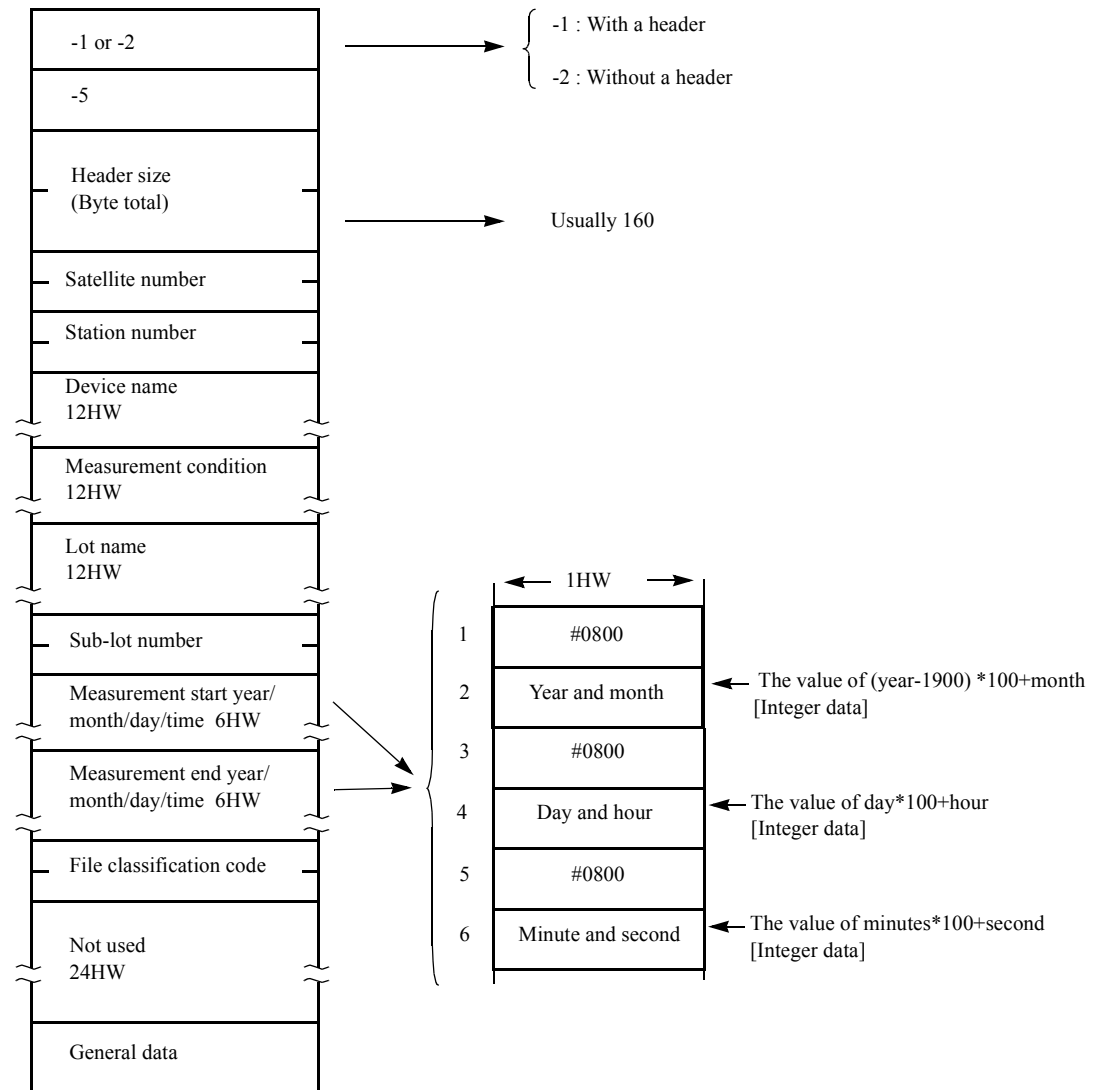


Tip

The specifications explained at (e), (f), and (g) are all header information.

Errors using the J command and the prime causes

- The specified channel is being used.
- There is a mistake in the file name.

Figure 2-1 Data File Header Structure

2.12 V Command

This command closes data files.

Pressing the [V] key from VKT will display the following message:

```
CLOSE DATA FILE!  
CHANNEL?  1↓      ←Specify the channel number of the channel to be closed.  
  
CLOSE OK!
```

No error is generated even if the associated channel has been opened with the channel number specified. (This command functions the same as the CLEAR! statement.)

More than one channel number can be specified. In this case, use a comma (,) or a space to separate the numbers.

Channel numbers must be specified using integers from 1 to 32.

2.13 .S Command

This command adds a sample number to a data file.

Entering the two characters, a period (.) and S from the VKT allows you to check or change the sample numbers specified for each station.

```
.S      ←Nothing is displayed.      ..... (a)
SPECIFY SAMPLE NUMBER STATION 1↓ ..... (b)
NUMBER? [1002] 1↓ ..... (c)
```

(a) Specify the command

Enter a period (.) and [S]. These input characters are not displayed when they are entered.

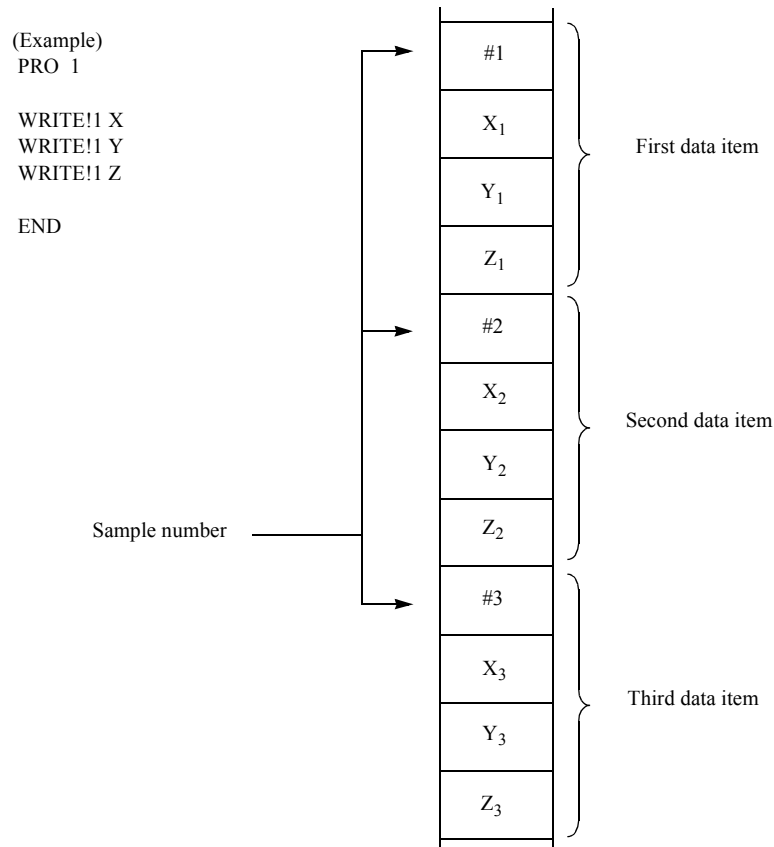
(b) Specify the station

This specifies the station number. Select 1 or 2.

(c) Specify the sample number

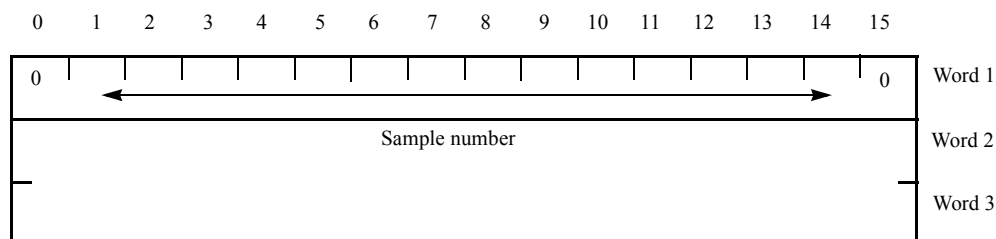
The number in [] brackets is the current sample number in decimal notation. If this sample number need not be changed, enter only ↓ (CR). If this sample number needs to be changed, enter the value and press ↓ (CR).

If the % command was specified for the data file at open processing, the sample number is written to the file the first time an attempt is made to write to the file during test program execution. In other words, the sample number is written to each device to be measured when the test program is executed during data file creation.



Tip

- The data file is assumed to be already open.
- The sample number is incremented when the test program terminates (i.e. test end).
- Sample numbers written into data files must be 3-word integer data as follows:



- *Sample numbers can be written to data files that have been opened by a J command with % or by an OPEN! statement with %.*
This processing can be prohibited by the action selector mode 8.
 - ➔ For more information on the action selector mode, refer to [2. 10 "Z Command" on page 2-36](#).
-

2.14 .V Command

This command initializes (INIT) and releases (RELEASE) devices.

The device or customer to be initialized (INIT) or released (RELEASE) is specified by entering a device identifier.

Pressing the [.V] key from VKT will display the following message:

```
.v  ←Nothing is displayed. .... (a)
INIT(0) OR RELEASE(1)?  n  .... (b)
WHAT DEVICE? d.s.↓  .... (c)
```

(a) Specify the command

Enter a period (.) and [V]. These input characters are not displayed when they are entered.

(b) Specify initialization (INIT) or release (RELEASE)

The system prompts to specify initialization (INIT) or release (RELEASE). Select 0 or 1. ↵ (CR code) is not necessary.

(c) Specify the device name or customer name to be initialized (INIT) or released (RELEASE)

Enter the name of the device or the customer to be initialized or released and terminate the program with ↵ (CR). The device name is specified as follows:

FP0, DP0, MT0, SERV

When SERV is entered as a device name, be sure to execute INIT SERV on CLI.

Tip

DP0 and SERV cannot be used as a device name in the ADVANsite systems.

Error messages are output as follows:

ERROR! AT INIT, RELEASE CODE=XXXXXX

Error code of 6 octal digits

➔ For more information on error codes, see the error code table.

2.15 .T Command

This command sets the test failure counter.

The test failure counter operates as described below.

- Set the test item number to be counted using this .T command.
- Second, the test program is executed. If measurement of the set test item fails, the counter is incremented by 1.

The counter value is incremented at the TEST END.

- The counter value can be referenced by the L command.

Pressing the [.T] key from VKT will display the following message:

```
TEST FAIL COUNTER
STATION?  1 ..... (a)
PRO?  1234↓ ..... (b)
STN-SPN1? 1-5<LF> ..... (c)
STN-SPN2? 20-29↓
```

(a) Specify the station number

Enter the station number.

Independent counter areas are provided for each station.

(b) Specify the program name

Enter the program name.

When the program actually executed is not the same as the program whose name is specified here, the counter is not incremented.

(c) Specify the test item number

Specify the start and end of the test item number to be counted and close the data file with ↓ (CR) or <LF> (line feed).

If it is closed with ↓, the setting ends. If it is closed with <LF>, additional settings can be specified. Up to 18 groups of settings can be specified.

The character n in STN-SPNn indicates the type number.

Tip

There is no particular limit to the total number of test items that can be set. But the area secured here is taken directly from the user program area.

Each counter requires two bytes. It does not result in an error if the same test item number is repeated. The only problem is that duplication requires more space, and this space may reduce the program area.

2.16 C Command

This command clears the measurement counter.

Entering this command will display the following message:

```
C CLEAR COUNTER!  
STATION NO.=
```

In response to this message, enter a station number (1 or 2) of the station to be cleared.

Enter **A** to clear both stations. This operation clears the following information:

- Total counter
- Failure counter
- Category counter
- Test fail counter
- Sort counter
- DC parametric test results

If the test fail counter has been specified by the .T command, the system outputs the following message. It prompts to confirm whether or not the counter setting should be cleared.

```
CLEAR TEST FAIL INFO. YES(1) OR NO(0)?
```

If 1 is entered in response to this question, the test fail counter is cleared. Enter 0 to not clear the test fail counter.



Tip

While the program is executed, the C command cannot be used to specify the station where SDUT exists.

2.17 L Command

This command lists out the measurement counter values.

Pressing the [L] key from VKT displays the following:

```
LIST UP COUNTERS COMMAND MENU
 1 : STATION1 DUTS COMBINED
 2 : STATION2 DUTS COMBINED
CA : ALL DUTS COMBINED
C1 : STATION1 DUTS COMBINED
C2 : STATION2 DUTS COMBINED
S1 : STATION1 DUTS SEPARATED
S2 : STATION2 DUTS SEPARATED
OF : OUTPUT FORMAT
COMMAND ? N
```

The command specified by N lists the contents of various counters.

The following counters are listed:

- Total/Pass/Fail counter
- Category counter
- Sort counter
- Test fail counter

The following contents are listed according to the command n values:

- When n = 1 or C1, the integrated values of all DUTs in station 1 are listed
- When n = 2 or C2, the integrated values of all DUTs in station 2 are listed
- When n = CA, the integrated values of all DUTs are listed
- When n = S1, the contents of all DUTs in station 1 are displayed separately
- When n = S2, the contents of all DUTs in station 2 are displayed separately

The contents of the test fail counter are output only when n = 1, 2, C1, C2, or CA.

When n = OF, the display formats of category counter, sort counter, and test fail counter in 1 to S2 above can be changed.

Select one of the following three formats:

ALL

Always displays category counter and sort counter.

For the test fail counter, displays all of the specification range of the .T command.

ALL can be selected by entering **AL**.

EXTRACT

Category counters are divided into blocks which have fifty counters (1 to 50, 51 to 100, ..., 900 to 950, and 951 to 999), and when all the counter values in a block are 0, the block is not displayed.

Sort counters are divided into blocks which have eight counters (1 to 8 and 9 to 16), and when all the counter values in a block are 0, the block is not displayed.

Test fail counter displays the value within the .T command specification range and with the count not 0.

EXTRACT2

Category counters and sort counters are not displayed when the data counter values in a single horizontal row are all 0.

Test fail counter displays the value within the .T command specification range and with the count not 0.

Enter **L**.

```
LIST UP COUNTERS COMMAND MENU
  1 : STATION1 DUTS COMBINED
  2 : STATION2 DUTS COMBINED
CA : ALL DUTS COMBINED
C1 : STATION1 DUTS COMBINED
C2 : STATION2 DUTS COMBINED
S1 : STATION1 DUTS SEPARATED
S2 : STATION2 DUTS SEPARATED
OF : OUTPUT FORMAT (ALL)      ← Displays the previously selected mode
COMMAND ? OF
SELECT OUTPUT FORMAT, ALL(AL) EXTRACT(EX) OR EXTRACT2(E2) ? EX

LIST UP COUNTERS COMMAND NEMU
  1 : STATION1 DUTS COMBINED
  2 : STATION2 DUTS COMBINED
CA : ALL DUTS COMBINED
C1 : STATION1 DUTS COMBINED
C2 : STATION2 DUTS COMBINED
S1 : STATION1 DUTS SEPARATED
S2 : STATION2 DUTS SEPARATED
OF : OUTPUT FORMAT (EXTRACT)
COMMAND ?
```

Output sample when n = CA

```

LIST UP COUNTER   AT 1998/03/04
DEVICE:  256K-DRAM  LOT NO.: 1-0                OPERATOR: TR
OUT I/F:                TESTER: 0   STATION: 1-2   DUT: 1-4
COMMENT:  L COMMAND

      TOTAL = 8      PASS = 6 (75.00%)      FAIL = 2 (25.00%)

CAT-001 CAT-002 CAT-003 CAT-004 CAT-005 CAT-006 CAT-007 CAT-008 CAT-009 CAT-010
      1       1       1       1       1       1       1       1       0       0
    12.50%   12.50%   12.50%   12.50%   12.50%   12.50%   12.50%   12.50%   0.000%   0.000%

CAT-011 CAT-012 CAT-013 CAT-014 CAT-015 CAT-016 CAT-017 CAT-018 CAT-019 CAT-020
      0       0       0       0       0       0       0       0       0       0
    0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%
      :       :       :       :       :       :       :       :       :       :
      :       :       :       :       :       :       :       :       :       :
CAT-181 CAT-182 CAT-183 CAT-184 CAT-185 CAT-186 CAT-187 CAT-188 CAT-189 CAT-190
      0       0       0       0       0       0       0       0       0       0
    0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%

CAT-191 CAT-192 CAT-193 CAT-194 CAT-195 CAT-196 CAT-197 CAT-198 CAT-199 CAT-200
      0       0       0       0       0       0       0       0       0       0
    0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%
      :       :       :       :       :       :       :       :       :       :
      :       :       :       :       :       :       :       :       :       :
CAT-981 CAT-982 CAT-983 CAT-984 CAT-985 CAT-986 CAT-987 CAT-988 CAT-989 CAT-990
      0       0       0       0       0       0       0       0       0       0
    0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%

CAT-991 CAT-992 CAT-993 CAT-994 CAT-995 CAT-996 CAT-997 CAT-998 CAT-999
      0       0       0       0       0       0       0       0       0
    0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%

SORT-01 SORT-02 SORT-03 SORT-04 SORT-05 SORT-06 SORT-07 SORT-08
      2       2       2       2       0       0       0       0
    25.00%   25.00%   25.00%   25.00%   0.000%   0.000%   0.000%   0.000%

SORT-09 SORT-10 SORT-11 SORT-12 SORT-13 SORT-14 SORT-15 SORT-16
      0       0       0       0       0       0       0       0
    0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%   0.000%

TEST FAIL COUNTERS OF PRO M
T-00000=000007  T-00001=000001  T-00002=000002  T-00003=000003  T-00004=000004

```

2.18 N Command

This command lists out the failed test numbers.

Pressing the **[N]** key from VKT displays the following:

```
LIST UP FAIL TEST NO.
COMMAND MENU
N   :   ALL DUTS
1   :   STATION1 DUTS COMBINED
2   :   STATION2 DUTS COMBINED
S1  :   STATION1 DUTS SEPARATED
S2  :   STATION2 DUTS SEPARATED
COMMAND N ?
      ↑
    Preceding setting
```

If **N** is entered, the test and sort numbers are displayed in the following format:

```
***** FAIL TEST DISPLAY *****      "P":PASS "*" :FAIL "-" :NO-TEST

TN0 DDDD
    0000   ← DUT number (hundreds)
    0000   ← DUT number (tens)
    1234   ← DUT number (ones)

001 PPPP
002 *PPP   ← PASS
003 -PPP
100 -PP*   ← FAIL
200 -P*-   ← NO-TEST
  ↑
Test number

SRT 0000   ← Sort number (tens)
    1231   ← Sort number (ones)
```

The following contents are listed when the commands 1, 2, S1, and S2 are entered respectively:

- When the command 1 is entered, the test numbers of failed tests are displayed for one or more DUTs in station 1.
- When the command 2 is entered, the test numbers of failed tests are displayed for one or more DUTs in station 2.
- When the command S1 is entered, the test numbers of failed tests are displayed for each arbitrary DUT in station 1.
- When the command S2 is entered, the test numbers of failed tests are displayed for each arbitrary DUT in station 2.

Output sample

```
DEVICE : SAMPLE1          SAMPLE: 57          LOT NO: 1-0
DUT I/F: T3993           TESTER: 0            OPERATOR: ADVANTEST
STATION: 1               DUT: 7
COMMENT: ASX/U-51 UTILITY
000000000001111111112222222223333333333444444444
01234567890123456789012345678901234567890123456789
0 -----*-----*-----*-----*-----
50 *-----*-----P-----*-----*-----
100 *-----*-----*-----*-----*-----
150 *-----*-----*-----*-----*-----
200 -----*-----*-----*-----*-----
250 -----*-----*-----*-----*-----
300 -----*-----*-----*-----*-----
350 -----*-----*-----*-----*-----
400 -----*-----*-----*-----*-----
450 -----*-----*-----*-----*-----
500 -----*-----*-----*-----*-----
550 -----*-----*-----*-----*-----
600 -----*-----*-----*-----*-----
650 -----*-----*-----*-----*-----
700 -----*-----*-----*-----*-----
750 -----*-----*-----*-----*-----
800 -----*-----*-----*-----*-----
850 *-----*-----P-----*-----*-----
900 *-----*-----*-----*-----*-----
950 *-----*-----*-----*-----*-----
      :
      :
9000 -----*-----*-----*-----*-----
9050 *-----*-----P-----*-----*-----
9100 *-----*-----*-----*-----*-----
9150 *-----*-----*-----*-----*-----
9200 -----*-----*-----*-----*-----
9250 -----*-----*-----*-----*-----
9300 -----*-----*-----*-----*-----
9350 -----*-----*-----*-----*-----
9400 -----*-----*-----*-----*-----
9450 -----*-----*-----*-----*-----
9500 -----*-----*-----*-----*-----
9550 -----*-----*-----*-----*-----
9600 -----*-----*-----*-----*-----
9650 -----*-----*-----*-----*-----
9700 -----*-----*-----*-----*-----
9750 -----*-----*-----*-----*-----
9800 -----*-----*-----*-----*-----
9850 *-----*-----P-----*-----*-----
9900 *-----*-----*-----*-----*-----
9950 *-----*-----*-----*-----*-----
```

2.19 A Command

This command selects the disk access check mode.

Entering this command will display the following message:

```
ACCESS CHECK OF DISK YES(1) OR NO(0)?
```

Entering **1** in response to this message sets the disk access check mode, and entering **0** sets the disk access non-check mode.

If disk access check mode is selected, the following message is output when the program is called from storage on a disk during test program execution:

```
DISK ACCESS CHECK!!  
SUBR-ABCD PRO -100102      ..... (a)  
TEST 155                  ..... (b)  
LINE1 -1590 [ABCD.X-1590-0000] ..... (c)  
LINE2 (&) -0              ..... (d)
```

(a) Displays the file names

The file name on the right indicates the location of the program just executed (PRO10000102), and the file name on the left indicates the current program (SUBR-ABCD).

(b) Displays the test number

Test number

(c) Displays the line number

This is the current line number of the program, source program name, and record number and line number in the source program.

(d) Displays which statement was being executed

If several statements were written on a line by using **&**, this indicates which statement was being executed.

2.20 T Command

This command sets the trigger conditions required for debugging.

Entering this command will display the following message:

For the T5591 and T5581

```

TRIGGER SELECTION !
STATION NO.= ----- (a)

                                -- MAIN --      -- SUB --
1. MPAT TRIGGER SELECTION MODE
   X-ADDRESS                    DISABLE         DISABLE
   Y-ADDRESS                    DISABLE         DISABLE
   PROGRAM COUNTER              DISABLE         DISABLE
   MUT COMMAND                  DISABLE         DISABLE
   D-FLAG                      DISABLE         DISABLE
   TIMING SET                   DISABLE         DISABLE
2. MPAT X-ADDRESS               #0              #0
3. MPAT Y-ADDRESS               #0              #0
4. MPAT PROGRAM COUNTER         #0              #0
5. MPAT MUT COMMAND             NONE            NONE
6. TIMING SET                   0               0

KEY-IN CODE OF ALPG (M:MAIN,S:SUB) ? ----- (b)

KEY-IN CODE OF TRIGGER THAT YOU WANT TO CHANGE !
[ (1-6) OR (P&CR) OR (P&1-6) OR (CR) ] ? ----- (c)

```

The current setting status is displayed for MAIN and SUB respectively.

The displayed status of PROGRAM COUNTER, D-FLAG, and TIMING SET is the same for both MAIN and SUB. (These settings are always common.)

For the T5592, T5571P, T5377, T5376, T5375, and T5371

```

TRIGGER SELECTION !
STATION NO.= ----- (a)

1. MPAT TRIGGER SELECTION MODE
    X-ADDRESS          DISABLE
    Y-ADDRESS          DISABLE
    PROGRAM COUNTER    DISABLE
    MUT COMMAND        DISABLE
    D-FLAG             DISABLE
    TIMING SET         DISABLE
2. MPAT X-ADDRESS      #0
3. MPAT Y-ADDRESS      #0
4. MPAT PROGRAM COUNTER #0
5. MPAT MUT COMMAND    NONE
6. TIMING SET          0

KEY-IN CODE OF TRIGGER THAT YOU WANT TO CHANGE !
[(1-6) OR (P&CR) OR (P&1-6) OR (CR)] ? ----- (C)

```

For the T5336 and T5311

```

TRIGGER SELECTION !
STATION NO.= 1 ----- (a)

1. MPAT TRIGGER SELECTION MODE
    X-ADDRESS          DISABLE
    Y-ADDRESS          DISABLE
    PROGRAM COUNTER    DISABLE
    MUT COMMAND        DISABLE
    D-FLAG             DISABLE
    TIMING SET         DISABLE
2. MPAT X-ADDRESS      #0
3. MPAT Y-ADDRESS      #0
4. MPAT PROGRAM COUNTER #0
5. MPAT MUT COMMAND    NONE
6. TIMING SET          0
7. STROBE SELECTION    0

KEY-IN CODE OF TRIGGER THAT YOU WANT TO CHANGE !
[(1-7) OR (P&CR) OR (P&1-7) OR (CR)] ? ----- (c)

```

(a) Specify the station number

Enter the station number (1 or 2) where the trigger data is to be set.

(b) Specify ALPG

Enter the ALPG, M (MAIN) or S (SUB) to be set.

(For the T5592, T5571P, T5377, T5376, T5375, T5371, T5336, and T5311, MAIN and SUB cannot be specified.)

When the T5581 2WAY interleaved mode or the T5591 4WAY interleaved mode is in process, the setting to SUB PG has no meaning. Only the specification to MAIN PG is valid.)

(c) Specify the trigger number

Enter the number of the trigger to be set. If P1toP6 (P1toP7 for the T5336 and T5311) are specified, the trigger number for the P command can be set. If only P is entered, the last set trigger number is set for the P command.

MPAT TRIGGER SELECTION MODE

This is the selection of the trigger mode.

If all conditions to which ENABLE is set are satisfied, a trigger signal is output.

X-ADDRESS - - - Detection according to the X address

Y-ADDRESS - - - Detection according to the Y address

PROGRAM COUNTER - - - Detection according to the PROGRAM COUNTER (PC)

MUT COMMAND - - - Detection according to the MUT signal

D-FLAG - - - Detection according to the D-FLAG. D-FLAG is set by the JZD command.

TIMING SET - - - Detection according to the timing set of TG

To set ENABLE or DISABLE for each item, enter E or D (the first letter of ENABLE or DISABLE).

Tip

Even if MAIN and SUB can be specified, the designation to PROGRAM COUNTER, D-FLAG, and TIMING SET are commonly performed to MAIN and SUB regardless of the specification of MAIN and SUB.

MPAT X-ADDRESS

Specifies the X address when the trigger detection is executed according to the X address.

The PC value range is as follows:

T5593, T5377, T5376 and T5375

#0 to #3FFFF

T5592, T5591, T5586, T5585, T5581, T5581H, T5581P, T5571P, T5371, T5336, and T5331

#0 to #FFFF

MPAT Y-ADDRESS

Specifies the Y address when the trigger detection is executed according to the Y address.

The PC value range is as follows:

T5593, T5377, T5376 and T5375

#0 to #3FFFF

T5592, T5591, T5586, T5585, T5581, T5581H, T5581P, T5571P, T5371, T5336, and T5331

#0 to #FFFF

MPAT PROGRAM COUNTER

Specifies a PC value when the trigger detection is executed according to the PROGRAM COUNTER.

The PC value range is as follows:

T5593, T5377, T5376 and T5375

#0 to #7FF

T5592, T5591, T5586, T5585, T5581, T5581H, T5581P, T5571P, and T5371

#0 to #3FF

T5336 and T5311

#0 to #1FF

Even if MAIN and SUB can be specified, this designation is commonly set to MAIN and SUB regardless of the specification of MAIN and SUB.

MPAT MUT COMMAND

Specifies a signal name when the trigger detection is executed according to the MUT signal.

One or more signal names shown below can be set.

T5593, T5377, T5376 and T5375

W, R, M1, M2, C0 to C23

T5592, T5591, T5586, T5585, T5581, T5581H, T5581P, T5571P, and T5371

W, R, M1, M2, C0 to C15

T5336 and T5311

W, R, M1, M2, C0 to C11

If CLEAR is entered, the previous setting is cleared.

If no name is specified, NONE is displayed.

Tip

The trigger detection is executed in the specified cycle between CYP9 to 16 when C15 is selected as the trigger data in the T5592 test system with the cycle palette 16 extended option installed.

TIMING SET

Specifies a timing set number when the trigger detection is executed according to the timing set.

Specify a number between TS1 to TS16.

Even if MAIN and SUB can be specified, this designation is commonly set to MAIN and SUB regardless of the specification of MAIN and SUB.

STROBE SELECTION

Selects a strobe output at the STRB terminal on the trigger box.

For the T5336 and T5311, 1 to 2 can be entered for the STROBE SELECTION.

The correspondence between STROBE SELECTION numbers and STRB or WSTRB numbers is as follows:

STROBE SELECTION	STRB number or WSTRB number
1	STRB1 or WSTRBL1
2	STRB2 or WSTRBT1

Entry of numerals

Values of X-ADDRESS, Y-ADDRESS and PROGRAM COUNTER are displayed in hexadecimal. If they are newly set, enter the value in the following form.

- Hexadecimal setting

#0000↵

- Decimal setting

0000↵

- Results when addition or subtraction is performed on the previous set values

+0000↵

-0000↵

+#0000↵

-#0000↵

2.21 P Command

This command sets the trigger condition. The trigger condition can be set only for the trigger having the same trigger number specified for the P command in part (c) of the section for the T command.

➔ For more information, refer to [2. 20 "T Command" on page 2-58](#).

Entering this command will display the following message:

```
TRIGGER SELECTION IN PAUSE MODE!  
STATION NO.=
```

In response to this message, enter a station number (1 or 2) of the station to be set. If the test program is being executed, however, the station need not be specified. The station at which the test program is being executed is set automatically.

Following this operation, the previous value and the message linked to the trigger number specified for the P command in part (c) of the section for the T command are displayed. After this, make settings in the same way as for the T command settings.

Tip

The following message indicates that no trigger number was set for the P command in the T command screen.

```
TRIGGER PAUSE MODE IS NOT SELECTED
```

2.22 CTRL/B Command

This command halts the test program and indicates at what point in the test program execution was aborted. Entering this command during test program execution will display the following message:

```
KEY-BREAK!! (3BF5) ..... (a)
SUBR-ABCD PRO-100402 ..... (b)
TEST-1001 ..... (c)
LINE1-1590 [ABCD.X-1590-0000] ..... (d)
LINE2 (&)-0 ..... (e)
STRIKE"CR"KEY!!
IF YOU WANT TO RESTART EXECUTION.
```

In response to this message, enter ↵ (CR) to continue the test program.

(a) Displays the memory address

(3BF5) is the memory address of the controller that was running when the CTRL/B command was accepted.

(b) Displays the file names

The file name on the right indicates the location of the program just executed (PRO100402). The file name on the left indicates the current program (SUBR-ABCD).

(c) Displays the test number

Test number

(d) Displays the program name

This indicates the current line number of the program, source program name, and record number and line number in the source program.

(e) Displays the statement

If several statement were written on a line by using &, this indicates which statement was being executed.

2.23 CTRL/T Command

This command traces the test program.

Entering this command will display the following message:

```
TRACE MODE SELECTION!
DISABLE (0) TRACE (1)
TRACE & PAUSE (2) : MODE=
```

In response to this message, enter 0, 1, or 2.

- **0** clears the trace mode.
- **1** sets the trace mode.
- **2** sets the trace mode and causes the test program to be halted temporarily each time a trace is run. If the HOLD switch on the control box is OFF at the time, the program will be placed in reference-off state. Tracing is performed each time a statement is executed.

The trace mode screen is as follows:

```
SUBR-ABCD PRO-100102 150 [ABCD.X-1-150]
(a)          (b)      (c)   (d)   (e)
```

- (a) Displays the program name
Name of the current program
- (b) Displays the program name
Name of the program just executed
- (c) Displays the line number of the program
Current line number of the program
- (d) Displays the source program name
Name of the source program of the current program
- (e) Displays the record number and line number
Source program record number and line number

2.24 U Command

This command executes the following:

- Clearing the user program area
- Forcibly clear data files
- Clear key commands
- Save the current variable area on a magnetic disk and recover the area
- Change the customer name
- Save a named protected memory on a magnetic disk

Entering this command will display the following message:

```

USER'S AREA COMMAND
0 : NO-OPERATE
1 : CLEAR SAME AS (2,4)
2 : CLEAR VARIABLE & PROGRAM AREA
3 : CLEAR DATA-FILE
4 : CLEAR KEY-COMMAND
5 : SAVE USER'S AREA TO "SYS CONT.BF"
6 : RECOVERY USER'S AREA FROM "SYS CONT.BF"
7 : LIST & CHANGE CUSTOMER NAME
8 : SAVE NAMED PROTECTION MEMORY
MODE NO.=

```

Enter a number (0 to 8) to perform the operation.

0 : NO-OPERATE

Nothing will be performed.

1 : CLEAR SAME AS (2,4)

This calls the operations 2 and 4.

2 : CLEAR VARIABLE & PROGRAM AREA

This opens the user program area and deletes the SYS CONT.BF file from this area.

- Clearing the user program area

Programs that are used frequently can be held in the user program area by the protect function to keep them available for processing.

This speeds up processing as long as there are not too many protected programs in the user program area.

If protected programs take up too much user program area, additional programs have to be retrieved from disk storage during execution, and this cuts down the execution speed.

If this happens, it may be advisable to delete all programs from the user program area.

- Deleting the SYS CONT.BF file

The SYS CONT.BF file saved by 5 in the message is deleted.

3 : CLEAR DATA-FILE

This closes the channel connected to the data file that remains open.

4 : CLEAR KEY-COMMAND

This initializes the following key commands:

- A
Disk access check
- F
FK mode selection
- I
Input device selection
- S
Output device selection
- M
DISABLE settings of SAMPLE DEVICE LOG, TEST NO. LOG, PIN NO. LOG, ALL DUT GO/NOGO and SET DUT NO.
- CTRL/T
Trace mode selection
- Breakpoints, insertion, deletion, etc. Set by the debugger.
- Data scan mode
- /BUFF
Name of the program to be executed when the program number is 0

5 : SAVE USER'S AREA TO "SYS CONT.BF"

This saves the following information in the SYS CONT.BF file on a magnetic disk:

- Data stored in the variable area
- Control box display data
- Trigger mode set by the T command
- TOTAL FAIL and CATEGORY counter values
- Test fail counter value
- FK mode set by the F command
- Channel information in the data file
- Operation mode set by the M command
- Breakpoints, insertion, deletion, etc. Set by the debugger.
- Data scan mode
- File name specified by the S command
- File name specified by the I command
- Sample number

- Test fail counter values specified by the .T command
- Values of the action selector mode and switch specified by the Z command

6 : RECOVERY USER'S AREA FROM "SYS CONT.BF"

This recovers the information saved above in 5. In this case, the information indicated in 5 becomes the information saved.

7 : LIST & CHANGE CUSTOMER NAME

This displays and changes the customer name. Entering this command outputs the following message:

```
CURRENT CUSTOMER XXXX
```

In response to this message, press the **↵** key if the customer is not going to be changed. If the customer name is to be changed, enter **customer name** and the **↵** key.

Do not enter the customer name SERV, which is reserved to identify the TESTARIUM or ADVANstar-21 service computer.

Tip

ADI\$DPI is displayed for CURRENT CUSTOMER on the ADVANsite systems. This is a logical name to stand for a current default directory.

8 : SAVE NAMED PROTECTION MEMORY

The memory secured using the hybrid controller basic system library AT_malloc_named() is saved on a magnetic disk. The file is created in the current customer directory (usually the directory in which the controller is started). The file name is the name given to the memory.

If the memory failed to be saved, the following message is output on VKT:

```
PROTECTION MEMORY DATA SAVE FAIL : XXXX
```

XXXX is the name given to the memory failed to be saved.

Memory blocks will not be released automatically regardless of whether the memory save is successful or fails.

➔ For more information on the content to be saved and how to recover it from a disk, refer to the AT_malloc_named() description in [ASX/U-51 Hybrid Controller Reference Manual\(No. : 8409600\)](#).

2.25 ESC, CTRL/E Commands

These commands pass control from the controller to the file monitor.

—  Tip —

ESC key has not been provided on some systems.

2.26 CTRL/I Command

T5500 series automatically executes the system initialize program (SUB INIT) 30 minutes after the power is turned on. The user can override automatic initialization by entering the CTRL/I command before that 30 minutes has elapsed. Entering this command will display the following message:

```
"INIT"MODE SELECTION, ENABLE(1) OR DISABLE(0)
```

If the user selects 0 (DISABLE) at this stage, automatic execution of the system initialization program is omitted. To execute the automatic system initialization, enter the CTRL/I command again, and select 1 (ENABLE).

If the user selects 1, automatic execution of the system initialization program is carried out.

When the system is activated, the ENABLE mode is on.

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